Inferential targets and the value of case volume in reproduction-number surveillance

Abstract

Estimates of the time-varying reproduction number are routinely interpreted as if additional reported cases always buy proportionally more precision. The return to case volume depends on the inferential target. Under a fixed, known reporting probability, a conditionally unbiased count estimate has mean squared error A/m about the current value, $A/(Tm)$ about the realised mean of T observed periods, and $v_R + A/m$ about a stationary process mean, where v_R is between-period transmission variance. Only the process-mean target retains an irreducible environmental term. If reporting completeness fluctuates independently and its current value is unknown, even current-value error retains a reporting variance floor, and the realised finite-period mean acquires a reporting term divided by T. Nonidentification of transmission and reporting variance from count moments follows directly.

We evaluate the empirical content of this distinction in US influenza hospital admissions over four early-growth seasons (1748 state-weeks). An affine working model gives an apparent balance ratio of 37.44 ([18.93, 69.92]), but held-out curve comparisons do not select a single form: the split is mixed for R_A, whereas both held-out summaries for G favour the power law. Controlled binomial thinning gives a counting-response rate of 0.4051 ([0.3656, 0.4413]); excluding the most extreme retention level raises it to 0.4848 ([0.4387, 0.5279]), above the approximate elasticity reference 0.4455. For R_A, state intercepts give a within-state ratio of 17.3 ([3.315, 47.8]), with a pooled-to-within paired ratio of 2.164 ([1.04, 7.745]) under calendar-week resampling; two- and three-week block intervals include one. These diagnostics are approximately consistent with the predicted scaling of counting noise while showing that cross-sectional surveillance does not identify a transmission-variance floor, a reporting mechanism or an operational case-volume threshold.

Keywords: reproduction number; environmental stochasticity; inferential targets; surveillance design; reporting heterogeneity; influenza hospitalisation

1. Introduction

The value of additional cases in reproduction-number surveillance depends on the quantity being estimated. A current transmission level, the average of transmission levels during an observed interval, and the mean of a process generating successive intervals are different inferential targets. Each can be relevant to surveillance, but they assign different roles to variation between periods. A change in transmission is part of the signal when estimating the current value; it contributes sampling variation when estimating a process mean. Statements about a precision limit therefore require both a target and an observation model. Case volume alone cannot determine that limit.

Reproduction numbers support epidemic assessment by summarising transmission under specified conditions (Fraser, 2007; Gostic et al., 2020). Renewal methods relate incidence to past infectiousness through a generation-interval distribution, and their uncertainty depends on the information in the observed counts (Cori et al., 2013; Thompson et al., 2019). Work on low incidence, reporting noise and timely detection has clarified important limits of period-specific inference (Parag, 2021; Parag et al., 2022, 2025). Temporal aggregation and delays create further differences between transmission and the apparent reproduction number of an observation series (Nash et al., 2023; Ogi-Gittins et al., 2024; Eales and Riley, 2024). These results motivate a distinction between uncertainty about a realised transmission trajectory and uncertainty about a population of periods from which that trajectory is sampled.

We develop that distinction in a one-generation branching experiment. Primary cases have independent negative binomial offspring, each secondary infection is reported independently, and the number of primary cases is fixed or exogenous. With a fixed, known reporting probability, a scaled count is conditionally unbiased for the current reproduction number. Its expected squared error decreases inversely with case volume. Averaging over a fixed set of observed periods similarly reduces counting error when the target is their realised mean. Estimating a stationary process mean adds the sampling variance of those periods. Increasing cases within the same periods leaves this additional term unchanged, whereas observing further periods can reduce it according to their dependence structure.

This distinction builds on a classical variance structure. Population ecology separates independent individual outcomes from environmental variation shared across a population (Engen et al., 1998; Lande et al., 2003); stochastic epidemic models likewise distinguish demographic and environmental contributions (Bretó et al., 2009; He et al., 2010). Shared conditions can include climate, contact patterns and school calendars (Cauchemez et al., 2008; Shaman and Kohn, 2009). In sampling theory, additional observations within a cluster reduce within-cluster error, while additional clusters address between-cluster variation (Cochran, 1977; Hayes and Bennett, 1999). The contribution here is to specify the analogous

terms for reproduction-number surveillance and show how changing the target changes the relevant risk. The resulting balance scale compares two contributions under a stated model; it is not a universal minimum case requirement.

Reporting requires separate attention. Independent omission of infections differs from a reporting probability that changes for all infections in the same period (Azmon et al., 2014; Stoner et al., 2019). Even a stable reporting probability can produce bias if its value is unknown or the count is incorrectly normalised. When reporting probabilities fluctuate independently of transmission, normalisation by their known mean leaves an additional error term for current-value estimation. It also affects the realised finite-period mean. For process-mean estimation, count means and variances alone generally cannot separate reporting volatility from transmission variation. Such ambiguity is consistent with the broader difficulty of separating process and observation variation in time series (Knape, 2008; Auger-Méthé et al., 2016).

Routine hospital surveillance does not directly observe the variables of the branching experiment. Admissions follow infection after a delay, aggregate heterogeneous populations and are recorded on a calendar schedule. Their count scales are outcomes of the epidemic, rather than exogenous design variables. We therefore examine squared relative deviations of apparent weekly growth statistics in US National Healthcare Safety Network influenza admissions (Centers for Disease Control and Prevention, 2026; Frutos et al., 2026). These deviations describe variation around an empirical centre; they are not squared errors against observed transmission truth. An affine association with inverse count can describe this variation without identifying its biological sources or establishing an asymptote outside the observed count range.

The empirical analyses distinguish questions that the data can address. Comparing affine and power-law curves concerns descriptive shape and prediction. Controlled binomial thinning examines the response to additional sampling of an observed trajectory. State-intercept fits examine within-state associations, while alternative centres assess sensitivity to common temporal patterns. Simulation connects these summaries to specified transmission and observation mechanisms, conditional on generation intervals, delays and dependence. None of these operations alone identifies environmental transmission variance. Their roles, uncertainty calculations and execution status are documented in Methods and Results (Sections 3 and 4).

The paper thus places inferential targets before case-volume interpretations. Section 2 states the exact identities and the moment nonidentification result, with the targets collected in Table 1. The empirical analysis then assesses associations and their sensitivity on the scale of the observed statistics. Proofs and symbolic extensions for forecasting, shrinkage and design appear in the appendix. Together, these components clarify when additional cases improve the chosen estimate and which remaining uncertainties require information beyond case counts.

2. Model and inferential targets

2.1. A branching experiment with a known observation scale

Consider one generation of transmission with $n \geq 1$ primary infections. Conditional on the reproduction number $R_t > 0$, primary case i produces independent negative binomial offspring (Lloyd-Smith et al., 2005),

$$X_{i,t} \mid R_t \sim \mathrm{NB}\left(k, \frac{k}{k+R_t}\right), \qquad \mathbb{E}(X_{i,t} \mid R_t) = R_t, \quad \mathrm{Var}(X_{i,t} \mid R_t) = R_t + \frac{R_t^2}{k}. \tag{1}$$

Here $k > 0$ controls individual heterogeneity. Each secondary infection is independently reported with probability $\rho \in (0, 1]$. Summation and binomial thinning give

$$\begin{aligned} C_t \mid n, R_t &\sim \mathrm{NB}\left(nk, \frac{k}{k+\rho R_t}\right), \\ \mathbb{E}(C_t \mid n, R_t) = \rho n R_t, &\qquad \mathrm{Var}(C_t \mid n, R_t) = \rho n R_t + \frac{\rho^2 n R_t^2}{k}. \end{aligned} \tag{2}$$

Define $m = \rho n$ and $\widehat{R}_t = C_t/m$. The design scale m is the expected reported count at $R_t = 1$, not the realised count C_t.

The baseline assumptions are fixed, known ρ and independent reporting (H1); conditionally independent offspring as in Eq. (1) (H2); fixed k, known for the likelihood calculation (H3); and a fixed or exogenous n independent of transmission (H4). Results are conditional on the selected design, with the same m in each period. Write $\mathbb{E}(R_t) = \bar{R} > 0$ and $\mathrm{Var}(R_t) = v_R < \infty$. Across repeated periods we additionally assume that counts are independent given the entire transmission trajectory, with conditional law (2). These are repeated branching experiments; they do not assert that naturally evolving incidence is exogenous.

This observation scale matters. Stability of an unknown reporting fraction does not establish unbiasedness: dividing by an assumed ρ_0 yields conditional mean $(\rho/\rho_0)R_t$. More cases do not remove the resulting multiplicative bias. A ratio constructed from noisy observed incidence also has a random denominator and need not share the properties of $C_t/(\rho n)$. The baseline therefore specifies an informative reference experiment, not a direct model for every surveillance statistic.

The conditional counting variance includes both offspring randomness and independent omission. Conditioning first on the total secondary infections separates their contributions as $(R_t + R_t^2/k)/n$ and $(1-\rho)R_t/(\rho n)$. Their sum equals the variance above. In particular, the term linear in R_t is not exclusively reporting noise: it persists with complete reporting. This distinction prevents a purely algebraic partition from being mistaken for a unique biological attribution.

2.2. Current values, realised means and process means

For the current target, $\mathbb{E}(\widehat{R}_t \mid R_t) = R_t$ and the conditional variance is $[R_t + \rho R_t^2/k]/m$. It attains the conditional Cramér–Rao bound in the one-parameter count model (Appendix Appendix A). The statement concerns estimators unbiased for every R_t under that likelihood. It is not a universal minimax bound, and it does not cover biased, hierarchical or prior-informed procedures.

Proposition 1 (One period, two targets)**.** *Under H1–H4, let $A = \bar{R} + \rho(\bar{R}^2 + v_R)/k$. Then*

$$\begin{aligned} \mathbb{E}[(\widehat{R}_t - R_t)^2] &= \frac{A}{m}, \\ V(m) := \mathbb{E}[(\widehat{R}_t - \bar{R})^2] &= v_R + \frac{A}{m}. \end{aligned} \tag{3}$$

Among estimators based on this period's count that are conditionally unbiased for R_t, $\widehat{R}_t$ minimises both risks.

The first risk averages the conditional estimation error over the stated distribution of R_t. It is not the conditional risk at every fixed value, nor a posterior credible interval for a particular period. The second evaluates the same observation against a different target. No assumption of independent transmission periods is needed for this single-period comparison, because only the marginal moments enter. Independence becomes relevant when risks are aggregated across periods.

The distinction follows from

$$\widehat{R}_t - \bar{R} = (\widehat{R}_t - R_t) + (R_t - \bar{R}). \tag{4}$$

Conditional unbiasedness makes the two terms uncorrelated. Between-period transmission variation is part of the current signal, but contributes sampling error for the process mean. The latter is a mean under a specified stationary law, not automatically a seasonal average in a nonstationary epidemic. The two contributions have the familiar within-unit and between-unit structure of clustered sampling (Cochran, 1977; Engen et al., 1998).

For $v_R > 0$, define the balance scale $m_\times = A/v_R$. Increasing m through n at fixed ρ, k and transmission law leaves A unchanged: the two terms in $V(m)$ are equal at $m = m_\times$. This scale describes the selected estimator and target. It does not mark a loss of all information, and changing ρ changes A as well as m. Coverage and budget calculations are given only as symbolic extensions in Appendix Appendix B.

Corollary 1 (A fixed set of observed periods)**.** *Under H1–H4 and conditional independence of counts given $R_{1:T}$, define $\bar{R}_T = T^{-1}\sum_{t=1}^{T} R_t$ and $\widehat{\bar{R}}_T = T^{-1}\sum_{t=1}^{T} \widehat{R}_t$. For integer $T \geq 1$, with common marginal moments of R_t,*

$$\begin{aligned} \mathbb{E}[(\widehat{\bar{R}}_T - \bar{R}_T)^2] &= \frac{A}{Tm}, \\ \mathbb{E}[(\widehat{\bar{R}}_T - \bar{R})^2] &= B_T + \frac{A}{Tm}, \qquad B_T = \mathrm{Var}(\bar{R}_T). \end{aligned} \tag{5}$$

For independent, identically distributed R_t, $B_T = v_R/T$. For a stationary process with $\mathrm{Cov}(R_t, R_{t+h}) = v_R\phi^{|h|}$, $|\phi| < 1$, the exact finite-T expression is $B_T = v_R S(\phi, T)/T^2$, where

$$S(\phi, T) = T + 2\sum_{h=1}^{T-1}(T-h)\phi^h = \frac{T(1+\phi)}{1-\phi} - \frac{2\phi(1-\phi^T)}{(1-\phi)^2}. \tag{6}$$

The realised-mean identity does not require independent transmission levels: only the counting errors must be conditionally independent, with the stated common moments. Increasing cases for fixed T removes its counting error, whereas process-mean risk approaches B_T. Additional periods address that term; dependence determines how much they contribute. Forecasting a future value is a further target, treated in Appendix Appendix B.

The stationary covariance formula uses integer periods and the complete finite sum, rather than a long-series approximation. Positive persistence usually reduces the gain from adjacent periods; negative autocorrelation can increase it. Neither a serially correlated process nor a season with changing mean should be replaced automatically by independent draws. For nonstationary transmission, the realised mean remains well defined, but interpretation of a common process mean requires an additional model.

Spatial averaging has an analogous condition. For J units with equal transmission variance v_R and mean pairwise correlation $\bar{c}$, the variance of their equally weighted transmission mean is

$$\mathrm{Var}\left(J^{-1}\sum_{j=1}^{J} R_{j,t}\right) = v_R\left[\bar{c} + \frac{1-\bar{c}}{J}\right]. \tag{7}$$

This covariance identity is not a decomposition of an empirical regression intercept. Unequal variances, unequal weights or a changing correlation structure require the corresponding covariance matrix.

Table 1. Inferential targets and exact mean squared errors. The fixed-reporting column assumes H1–H4. The volatile-reporting column assumes H2–H5 and, for averages, independent reporting probabilities independent of the entire transmission path. Counting errors are conditionally independent across periods. $B_T = \mathrm{Var}(\bar{R}_T)$ equals v_R/T for independent, identically distributed transmission and $v_R S(\phi, T)/T^2$ under the stated stationary autocovariance.

Target	Estimator	Fixed, known reporting	Unobserved volatile reporting
Current R_t	$\widehat{R}_t$	A/m	$v_H + A_H/m$
Realised mean $\bar{R}_T$	$\widehat{\bar{R}}_T$	$A/(Tm)$	$(v_H + A_H/m)/T$
Process mean $\bar{R}$	$\widehat{\bar{R}}_T$	$B_T + A/(Tm)$	$B_T + (v_H + A_H/m)/T$

2.3. Unobserved reporting volatility and identification

Replace H1 by a common period-specific probability $p_t \in [0, 1]$, independent of R_t and the design, with known mean $\rho > 0$ and $\mathrm{Var}(p_t) = \rho^2\tau^2$ (H5). Reporting remains independent across infections conditional on p_t. Necessarily $\tau^2 \le (1-\rho)/\rho$. The estimator still divides by $m = \rho n$; it does not observe p_t.

Proposition 2 (Reporting volatility and moment nonidentification)**.** *Under H2–H5, put $Q = \bar{R}^2 + v_R$. Then*

$$\begin{aligned} &\mathbb{E}(\widehat{R}_t) = \bar{R}, \qquad V_H(m) := \mathrm{Var}(\widehat{R}_t) = v_R + v_H + \frac{A_H}{m}, \\ &v_H = \tau^2 Q, \qquad A_H = \bar{R} + \frac{\rho(1+\tau^2)Q}{k}. \end{aligned} \tag{8}$$

For known m, ρ and k, the first two count moments depend on v_R and τ^2 only through $U = (1+\tau^2)(\bar{R}^2 + v_R)$. They therefore do not generally identify transmission variance and reporting volatility separately.

Conditional on both R_t and p_t, the estimator's mean is $(p_t/\rho)R_t$. Unbiasedness for R_t is restored only after averaging over independent reporting with the correct mean. This explains why more infections within a period cannot average away a shared reporting shock. If that period's reporting probability were measured and positive, a correctly adjusted estimator would pose a different problem; uncertainty in measuring that probability would itself require propagation.

This is a moment nonidentification result within the specified family. It does not say that external reporting measurements, temporal information under additional assumptions, or a more restrictive full distribution could never identify the mechanisms. Boundary cases can also impose further restrictions. The independence of reporting and transmission is substantive; if illness pressure changes reporting practice, the displayed decomposition need not hold.

Corollary 2 (Current and realised-mean risks with volatile reporting)**.** *Under H2–H5,*

$$\mathbb{E}[(\widehat{R}_t - R_t)^2] = v_H + \frac{A_H}{m}. \tag{9}$$

If, in addition, the p_t are independent and identically distributed, independent of the entire transmission trajectory, and counts are independent conditional on both trajectories, then

$$\mathbb{E}[(\widehat{\bar{R}}_T - \bar{R}_T)^2] = \frac{v_H + A_H/m}{T}, \qquad \mathbb{E}[(\widehat{\bar{R}}_T - \bar{R})^2] = B_T + \frac{v_H + A_H/m}{T}. \tag{10}$$

Thus, absence of a transmission-variation term for the realised target does not imply absence of reporting error. At fixed T, unobserved reporting volatility remains as m grows. Serially dependent reporting adds covariance terms, and a stable unknown mean reporting fraction can add bias. Table 1 collects the identities and their conditions.

2.4. Endogenous count scales and apparent statistics

If $m = \rho n$ is random and associated with transmission, even retaining the conditional count law and unbiasedness given (R_t, m) gives

$$\begin{aligned} \mathbb{E}[(\widehat{R}_t - \bar{R})^2 \mid m] = \frac{\mathbb{E}(R_t \mid m) + \rho\mathbb{E}(R_t^2 \mid m)/k}{m} \\ + \mathrm{Var}(R_t \mid m) + [\mathbb{E}(R_t \mid m) - \bar{R}]^2. \end{aligned} \tag{11}$$

These conditional moments can change with scale, so a constant intercept plus a constant inverse-count coefficient does not follow. Conditioning on an observed outcome count is still more restrictive: it need not preserve the assumed conditional observation law. A renewal representation alone does not resolve either problem.

The empirical response is consequently defined as squared relative deviation of an apparent statistic from its specified centre (Section 3.2), not error against known R_t. Its affine intercept and slope need not equal $v_R/\bar{R}^2$ and $A/\bar{R}^2$. Delays, aggregation, denominator noise and changing conditional means all affect that mapping. Calibration specifies a model-dependent connection (Section 3.6); it does not turn the empirical intercept into an exact additive decomposition of biological sources.

3. Materials and methods

3.1. Data, windows and the empirical target

We used the National Healthcare Safety Network (NHSN) weekly hospital respiratory-data snapshot retrieved on 20 October 2026, restricting influenza admissions to the 50 US states and the District of Columbia (Centers for Disease Control and Prevention, 2026; Frutos et al., 2026). National and regional aggregates and territories were excluded. Hospital participation varied during the series, including the voluntary-reporting interval in May–October 2024 and the subsequent reporting requirement (Centers for Medicare & Medicaid Services, 2024). The analysis is descriptive of this surveillance system and does not estimate an effect of that policy change.

The primary windows were the previously disclosed early-growth phases of four seasons: 1 October–26 November 2022, 7 October–23 December 2023, 9 November–28 December 2024 and 4 October–27 December 2025. Each comprises the contiguous national-growth segment with weekly growth at least 1.05 preceding the first turning point after the season's admissible start. The 2024 start follows resumed mandatory reporting. This rule and the primary statistic had been selected in earlier analysis rounds; the present revision is not a preregistered study. We preserved the windows rather than selecting them again to strengthen a fitted intercept. Inference conditions on their selection.

We rebuilt observations from the raw snapshot, preserving missing or suppressed counts as missing. A row requires four observed counts on consecutive weekly dates, from two weeks before its denominator week through the next week. Missing weeks are not treated as zero admissions. We retained genuine zero counts. The weighted scale must satisfy $D_t \geq 5$; the primary single-week statistic also requires $C_t \geq 5$. The resulting samples contain 1748 and 1756 state-weeks for R_A and G, respectively, across 51 jurisdictions and 42 calendar weeks. The difference between samples is determined by these denominator conditions. State-week keys and original counts are frozen before any thinning analysis.

3.2. Apparent statistics and relative squared deviation

Let C_t denote admissions in week t. We used the weighted growth statistic

$$G_t = \frac{C_{t+1}}{D_t}, \qquad D_t = 0.65C_t + 0.25C_{t-1} + 0.10C_{t-2}, \tag{12}$$

and the growth-to-reproduction transform

$$R_{A,t} = \left[1 + \frac{\theta}{7} \log \frac{C_{t+1} + 0.5}{C_t + 0.5}\right]_+^{\alpha}, \qquad \alpha = (g_\mu/g_\sigma)^2, \quad \theta = g_\sigma^2/g_\mu. \tag{13}$$

The positive-part convention defines the transformation when its base is nonpositive. It is a numerical convention, not an assertion that the exponential-growth approximation holds there. The gamma-interval mapping follows the renewal relationship between growth and reproduction (Wallinga and Lipsitch, 2007). Neither statistic is an observed true reproduction number or a posterior estimate from an infection-renewal model (Cori et al., 2013; Eales and Riley, 2024).

For continuity, the primary transform uses $g_\mu = 3.6$ days and $g_\sigma = 1.6$ days, the household serial-interval estimate of Cowling et al. (2009). We distinguish this proxy from a generation interval. Sensitivities use mean 3.2 days with the original coefficient of variation held fixed, and mean 3.2 days with SD 2.1 days, the intrinsic generation-time estimates in Table 2 of Chan et al. (2025). The fixed-coefficient-of-variation choice is an assumption. The reported household realised generation time and subtype serial intervals are different quantities (Chan et al., 2025; Vink et al., 2014).

For each specification, we calculated

$$L_{st} = \left(\frac{X_{st} - \bar{X}}{\bar{X}}\right)^2, \qquad E(L_{st} \mid D_{st}) = a + b/D_{st}, \tag{14}$$

where X is an apparent statistic and $\bar{X}$ is its pooled mean. For R_A, C_t replaces D_t in the regression. This response measures dispersion around a fitted reference; it is not squared estimation error against known R_t. The ratio b/a, when defined, describes the affine working model on that statistic's count scale. It is not directly the latent balance scale $m_\times$.

3.3. Fitting, uncertainty and model comparison

The affine mean was fitted by iterated feasible generalised least squares (FGLS), with precision proportional to the inverse squared fitted mean. Intercepts and slopes can be signed, while fitted means on observed rows must remain positive. Damped updates decrease the gamma quasi-deviance objective $\sum_i w_i[L_i/\mu_i + \log \mu_i]$; a positive-mean numerical fallback uses the same objective. We record convergence, rank and failures. Reaching an iteration limit does not produce a reported estimate. This objective is an estimating criterion, not a claim that the empirical losses follow a gamma likelihood.

Primary pooled intervals use 2000 state $\times$ calendar-week pigeonhole bootstrap draws (Owen, 2007). A row receives the product of its independently drawn state and week multiplicities. The reference mean and regression are re-estimated in each draw. Secondary analyses use 1000 draws. We also resample two- and three-week blocks within seasons, combined with state multiplicities, to examine sensitivity to persistence in national shocks. These few seasons do not support

unrestricted inference across arbitrary future epidemic regimes. Undefined ratios, denominator signs and failed fits are retained in diagnostics; any finite-draw interval is labelled accordingly.

We compared the affine mean with $\lambda D^{-\gamma}$ by leaving out one state or one season at a time. All response centres are estimated from the training observations; held-out losses use that training centre. Each model is fitted only on its training fold. Held-out scores are the mean of $L/\mu + \log \mu$, with squared prediction errors retained as a secondary measure. We report power-law minus affine scores, so a positive difference favours the affine model. The overlapping training folds are not independent replications, and four season folds are not used to manufacture a small-sample significance test. The earlier residual-bootstrap platform p-value is not carried forward as a corrected test.

3.4. Controlled binomial thinning

At $q \in \{0.8, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1\}$, each unique state-week count is sampled once from $\mathrm{Bin}(C, q)$ per realisation and reused in every overlapping ratio. No eligibility filter or window is reapplied. We recompute the statistic, its centre and the affine fit using the original C_t as regressor scale. The point response averages 400 separately fitted realisations. Each of 1000 outer bootstrap draws averages 50 inner fits per level. The point estimate therefore averages 400 realisations, whereas each bootstrap draw averages 50. The interval consequently includes finite-inner simulation variation in addition to sampling variation; the Monte Carlo diagnostics quantify both components rather than removing the former. The precision rule, recorded before results were reported, requires the Monte Carlo standard error of each of the five primary response summaries to be at most 10% of its point estimate; if not, inner counts are doubled uniformly up to a cap, and the actual budget and remaining simulation error are reported. A criterion relative to bootstrap sampling error was rejected because that ratio grows with the outer budget and cannot be met at any feasible inner budget (Section efapp:inputs).

The approximate relation is

$$b(q) - b(1) \simeq c_{\text{thin}}(1/q - 1). \tag{15}$$

We fit the through-origin rate, retain each level and repeat the summary without $q = 0.1$ and with each level omitted in turn. Prespecified subsets require original $C_t \geq 50$, or both $C_t, C_{t+1} \geq 50$. These conditions bound expected thinned counts, not every random realisation. A reference coefficient is the weighted mean of $\kappa(g)^2[1 + (C_t + 0.5)/(C_{t+1} + 0.5)]$, where $\kappa(g) = (g_\mu/7)/[1 + \theta \log(g)/7]$ and $g = (C_{t+1} + 0.5)/(C_t + 0.5)$. It approximates the relative transformation response and stabilises inverse counts; it is not the exact variance of the recentered nonlinear statistic. We recompute this coefficient in every bootstrap draw to obtain paired c_{thin}/b_P intervals. Simultaneous bands describe the vector of intercept changes. A known Poisson-count example runs through the same procedure as a mechanism check, not as a coverage validation.

3.5. State heterogeneity and reference sensitivity

With the same pooled-loss definition, we fit $E(L_{st} \mid C_{st}) = a_s + b_{\text{w}}/C_{st}$. State summaries target sampled state occurrences: repeated draws of a state count repeatedly, and absent states do not enter as zero coefficients. Row- and case-weighted summaries use resampled totals. Pooled and within-state contrasts are paired within draws. We separately replace C_t by the actual weighted D_t. Week-only and state-plus-week fits are numerical diagnostics; failed fits are explicitly unavailable.

A separate analysis changes the reference in the loss, using season means, season-specific linear trends common to states, national-week means, and leave-own-state-out national-week means. The global mean squared remains the denominator, isolating centring from renormalisation. All centres and trends are refitted under bootstrap weights. These operations change the empirical target and may remove real transmission signal. Their fitted intercept differences are therefore specification sensitivities, not orthogonal biological variance components. We additionally report $\mathrm{Var}_w(\bar{d}_w)/\mathrm{Var}_{st}(d_{st})$ for signed deviations $d = X/\bar{X} - 1$, explicitly as a descriptive variance ratio rather than an ICC of squared losses.

3.6. Conditional calibration and design illustrations

A daily renewal simulator reproduces the four fixed window lengths and initial state counts. Transmission has independent negative-binomial offspring; weekly reproduction-number shocks have separate national and state components with declared marginal variance and persistence. Admission delays and report probabilities are assumptions unless documented externally. Binomial admissions are allocated over delays. An alternative observation model samples directly from the latent delayed admission intensity λ, with conditional mean λ and variance $\varphi\lambda$; it does not resample an already noisy admission count. The Poisson limit is $\varphi = 1$. Observation-moment tests check these definitions independently.

We apply the same statistics, filters and fitting functions to simulated panels. The variance parameter describes weekly reproduction shocks within this simulator. It is not converted to a per-generation variance or an operational weekly case threshold. Paired empirical-intercept draws and paired simulated-panel resampling propagate sampling and calibration Monte Carlo uncertainty. Separate generation-interval and observation-model scenarios describe structural sensitivity; they are not a posterior distribution or a joint confidence interval over mechanisms. Supplementary methods specify all settings, boundaries and simulation diagnostics. Further supplementary analyses report season and participation sensitivities and an explicitly hypothetical curve-discrimination design experiment. None is used to infer a causal reporting-mandate effect or a universal surveillance sample-size requirement.

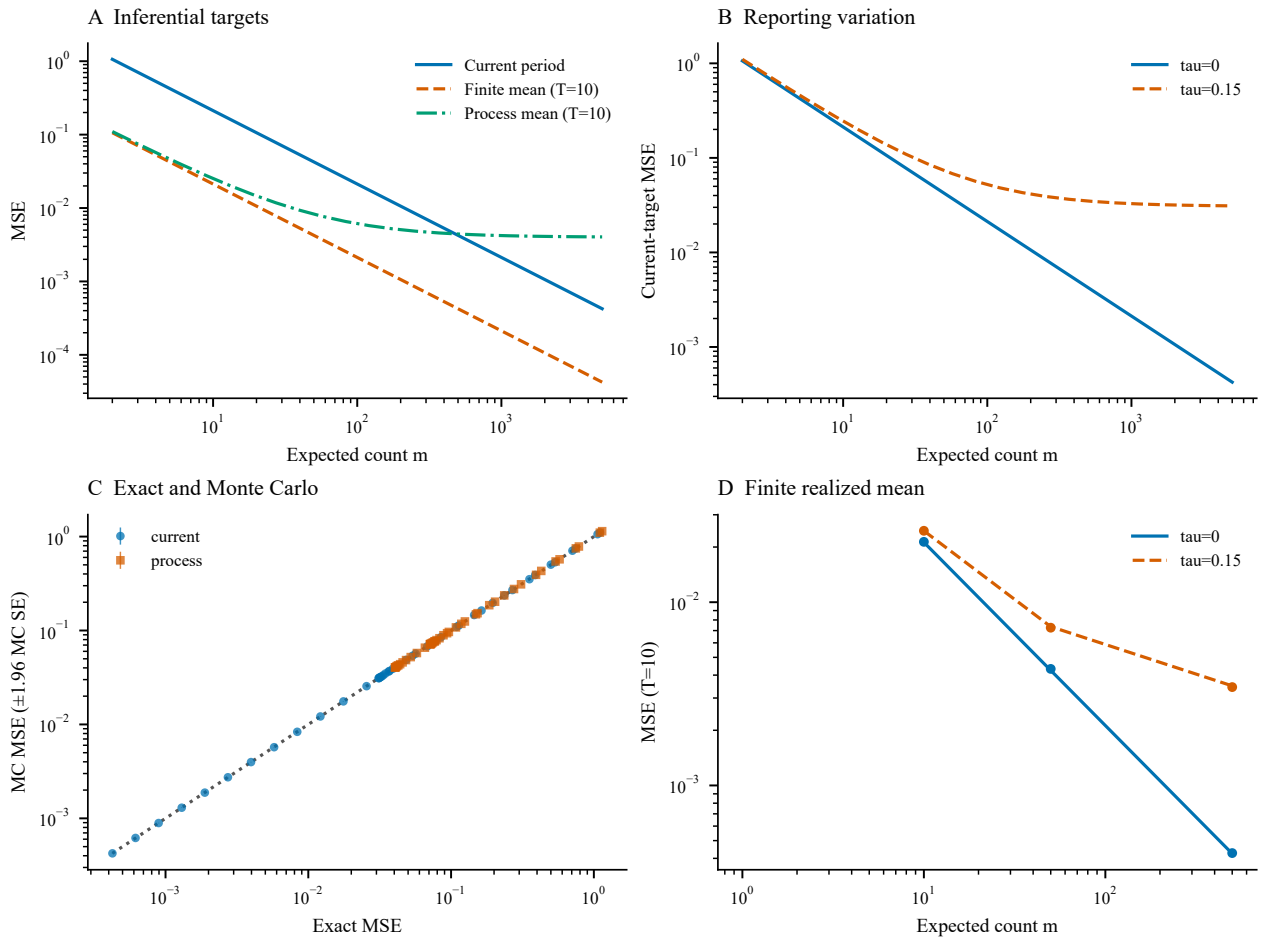


Figure 1. Inferential targets and reporting assumptions. Curves give the model risks and points, where shown, give known-truth Monte Carlo checks. Current-value and realised finite-period targets differ from the stationary process mean. Unobserved period-level reporting variation can leave a nonzero risk even for current values. Counts are generated under the stated fixed-exposure model; the figure does not identify a biological variance floor in the hospital data.

4. Results

4.1. Target-specific risks under known truth

The known-truth simulation reproduced the qualitative separation implied by the model (Fig. 1). With a fixed, correctly known reporting probability, error about the current reproduction number decreased with inverse case volume. Error about a fixed collection of realised reproduction numbers also decreased, because those realised fluctuations were part of the target rather than an error component. At a fixed number of sampled periods, error about the stationary process mean retained the variance of the realised mean. The simulation checks these formulas under their own assumptions; it is not evidence that an empirical admission-statistic intercept has the same origin.

When the estimate used the mean reporting probability while period-specific completeness fluctuated independently, current-value error approached the reporting term rather than zero. The finite-period mean inherited a reporting contribution divided by the number of independent periods. Thus the absence of an environmental floor for a realised target did not imply the absence of an observation-process floor. Simulation records include Monte Carlo standard errors for each comparison, so numerical differences are judged against simulation uncertainty rather than an arbitrary percentage agreement. The variance balance and budget examples in Appendix Appendix B refer to this specified model, not to operational thresholds inferred from admissions.

4.2. Observed loss curves and held-out comparison

The rebuilt primary R_A sample contained 1748 state-weeks. The affine working fit gave $a = 0.01495$ (95% interval [0.01001, 0.02092]), $b = 0.56$ ([0.369, 0.7692]) and an apparent ratio $b/a = 37.44$ ([18.93, 69.92]). The corresponding ratio for G was 23.46 ([8.279, 54.7]; Table S1). These are estimates of the apparent statistics' relative dispersion under the specified affine form. A positive conditional intercept does not establish the form of the curve outside the observed count range.

The two fitted curves described similar broad changes over the observed support, but their held-out rankings were not uniform (Fig. 2; Table S2). For R_A, the average power-law-minus-affine score was -0.002018 when leaving out states

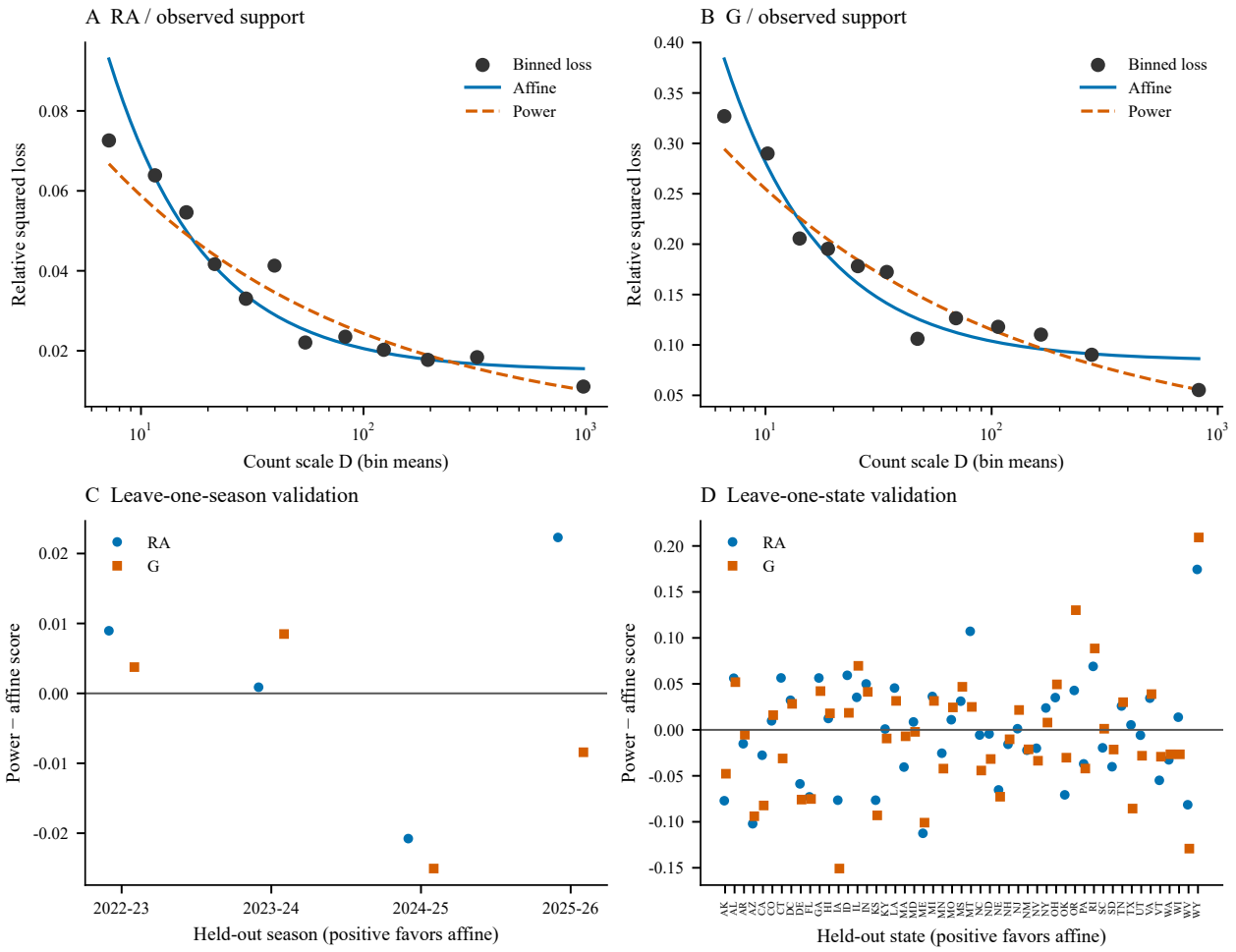


Figure 2. Empirical support and held-out curve comparison. Binned means describe relative loss over observed case counts; they are not independent observations and are not used as an unclustered significance test. The affine and power-law curves are fitted working means. Held-out score differences use centres and model parameters estimated within training folds. Positive power-law-minus-affine differences favour the affine model; negative differences favour the power law. Fold scores do not constitute independent replications.

and 0.004626 when leaving out seasons. The negative state score favoured the power law, while the positive season score favoured the affine model. For G, both summaries favoured the power law (-0.007369 and -0.004414). These comparisons now use training-fold centres throughout. Individual fold values are provided because a small aggregate difference conceals variation between held-out settings, and the four season folds supply little information about generalisation to new epidemic regimes.

Neither the fitted positive intercept nor this mixed predictive comparison provides independent evidence for an invariant transmission-variance floor. We therefore retain b/a only as an affine descriptive ratio. The 3.2-day transform sensitivities in Table S1 preserve the distinction between a changed apparent statistic and a change in the true interval assumed by a simulator. They do not identify which interval generated the observed admissions. The supplementary design experiment evaluates distinguishability under declared synthetic curves; its high power in some settings is not substituted for an empirical test, and its held-out null rejection rates are reported alongside power.

4.3. Response to controlled removal of cases

For the R_A sample (n=1748 state-weeks), across all retention levels the fitted thinning rate was 0.4051 (95% interval [0.3656, 0.4413]), against the approximate elasticity reference 0.4455. Its paired ratio to that reference was 0.9093 ([0.8101, 0.9995]). Excluding the most extreme retention level gave 0.4848 ([0.4387, 0.5279]; Fig. 3, Table S3). The contrast between these summaries matters because the $q = 0.1$ point supplies about 76% of the through-origin predictor sum of squares. This leverage is a property of the chosen inverse-retention design, not by itself proof that the observation is biased.

The original-$C_t \geq 50$ subset gave 0.4734 ([0.409, 0.5503]), while requiring both current and next counts to be at least 50 gave 0.4926 ([0.42, 0.5742]). The latter restriction addresses a limitation of the first: a large current count does not ensure a large numerator after thinning. All these subsets were defined from original counts, with no post-thinning reselection. Their different reference coefficients are recomputed rather than borrowing the full-sample reference.

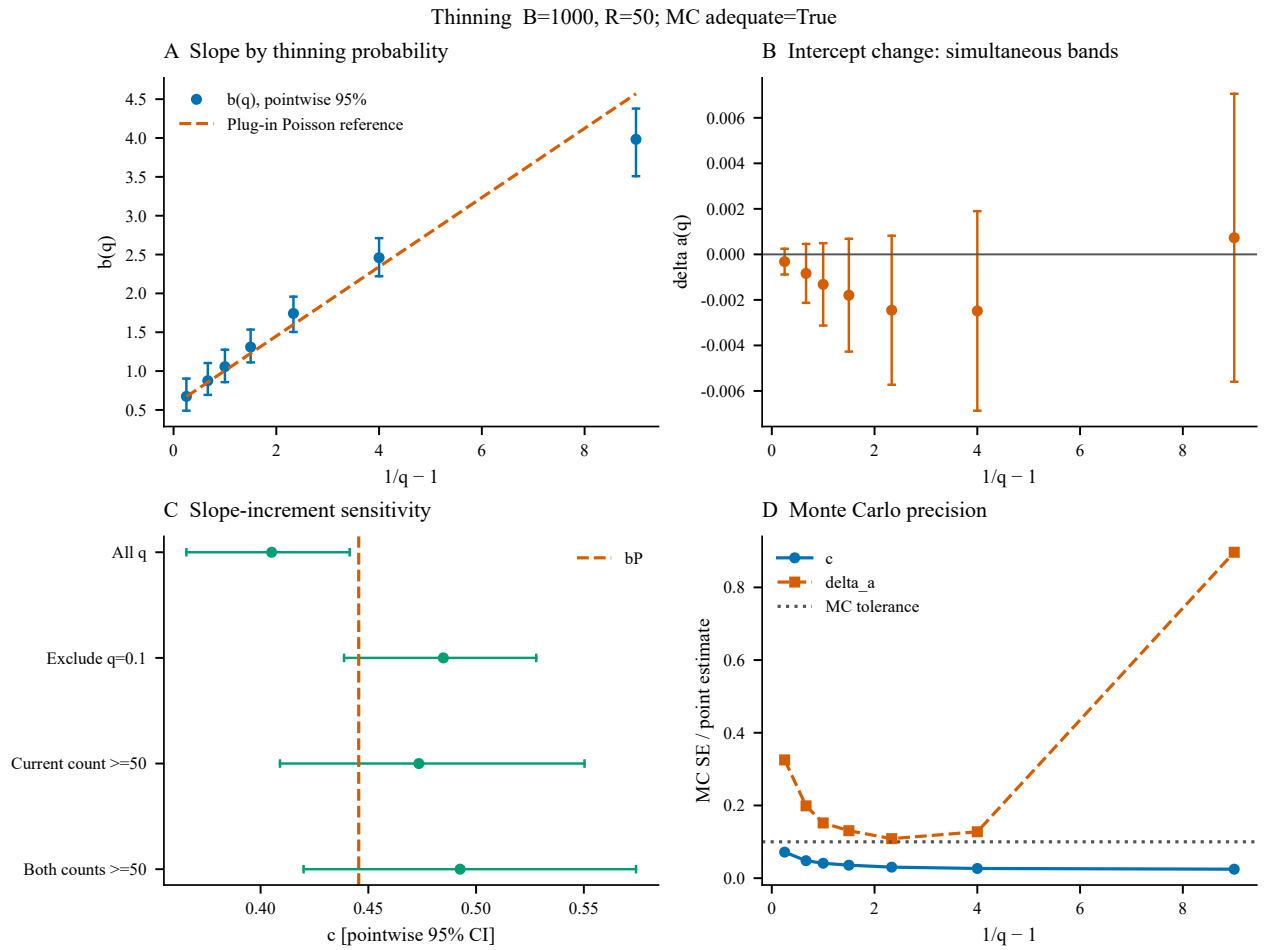


Figure 3. Controlled thinning on the frozen sample. Coefficients are fitted on the original count scale after recomputing the statistic and its centre. Intercept-change bands account for their joint bootstrap variation across retention levels. Sensitivity summaries retain all levels, omit $q = 0.1$, or restrict the original current and next counts. The Poisson elasticity coefficient is an approximate reference, not an exact prediction for every nonlinear fitted coefficient. Actual inner and outer repetition counts are reported in Table S3.

Figure 3 also shows changes in intercept and their joint uncertainty. These changes, including any departure at low retention, are part of the result. A confidence interval containing zero is not evidence of exact intercept invariance. The nonlinear transformation, continuity correction, recentering and refitting can modify the response even when the imposed count-removal mechanism is known. The known Poisson-count check in the supplementary output examines that complete pipeline, without claiming to represent every reporting mechanism in practice.

The calculation used 1000 outer samples, 400 point realisations and 50 inner realisations per bootstrap level. Numerical precision is documented in Table S3, including the predefined Monte Carlo check and fit failures. Because the point estimate averages more realisations than each bootstrap draw, the interval includes both sampling and finite-inner simulation variation; Table eftab:thinning reports the two components separately. The experiment establishes how these recorded counts respond to additional random removal. It does not separate transmission, admission practice, delayed observation and reporting variation in the original intercept, nor does the difference between $b(1)$ and c_{thin} uniquely quantify offspring overdispersion.

4.4. State heterogeneity and the choice of loss reference

For the same R_A sample, with state intercepts and a shared within-state slope, $b_w = 0.3794$ ([0.1082, 0.6403]) and the mean state intercept was 0.02192 ([0.01289, 0.03502]). The mean-based ratio was 17.3 ([3.315, 47.8]), and replacing the scale by the actual weighted D_t gave 17.71 ([3.267, 47.81]; Fig. 4, Table S4). The latter is a distinct fit, not an alias of the single-week count analysis.

The pooled-to-within-state paired ratio was 2.164 ([1.04, 7.745]) under independent resampling of calendar weeks. The two- and three-week block intervals were [0.964, 8.483] and [0.9866, 8.87], respectively. The ordinary interval excluded one narrowly, whereas the time-block intervals included it. Consequently, the apparent difference is sensitive to how temporal dependence is represented. It cannot be translated into a causal percentage attributable to spatial averaging. The primary state model supplied 999 valid fits among 1000 bootstrap samples; the conditional nature of finite-fit intervals and the diagnostic failures are explicit. The state-plus-week point diagnostic did not meet the numerical convergence criterion and is not assigned a slope or confidence interval.

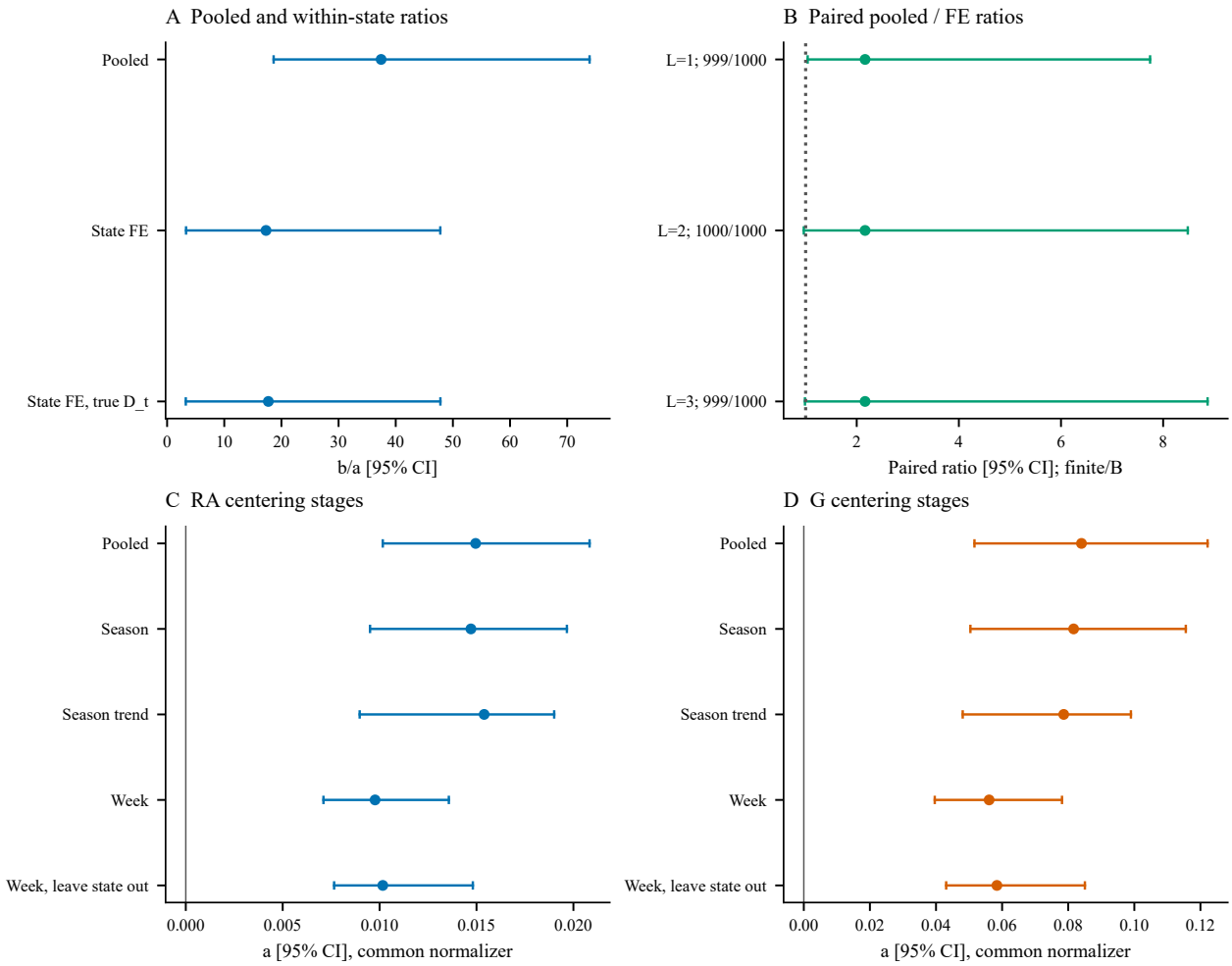


Figure 4. Spatial specification and reference sensitivity. Pooled and state-intercept estimates use the same underlying global-loss definition; the D_t sensitivity uses the weighted lagged count. Reference adjustments hold the global normalisation fixed and change the centre or common trend. Time-block paired intervals show sensitivity to the treatment of temporal dependence. Undefined or unconverged quantities are not drawn as successful estimates. Different references define different empirical targets.

Changing the reference of the loss also changed the intercept (Table S5). With the common global denominator retained, replacing the pooled centre by the national-week centre reduced the R_A intercept by 34.7% ([5.97, 51.9]%) and the G intercept by 33.3% ([6.8, 51.6]%). The reduction is specific to holding the denominator fixed; changing the normalisation as well would define a different, non-comparable quantity. Season centring and common linear detrending did not reproduce that same change. This is expected: different references remove different components of the observed trajectory, including genuine common signal. The leave-own-state-out variant documents the influence of using an observation in its own weekly reference. These comparisons describe specification sensitivity, not an identified decomposition into independent transmission and reporting variances.

4.5. Limits of the conditional calibration

The supplementary calibration applies the corrected observation mechanism to both apparent statistics and reports all declared scenarios (Table S6). It distinguishes the source-matched 3.2-day, SD 2.1-day interval from a fixed-coefficient-of-variation sensitivity and the historical 3.6-day proxy. Conditional intervals combine paired empirical and simulator uncertainty; they do not average over an invented distribution of interval assumptions.

A simulator can reproduce an intercept while failing to reproduce the slope or retained sample size. Those discrepancies are therefore retained in the response checks rather than eliminated by adjusting initial zero counts or selecting a preferred scenario. The initial-condition and no-importation assumptions can leave initially zero states without subsequent infections, which differs from surveillance settings with continuing introductions. National and state persistence, count overdispersion and reporting volatility are likewise separate mechanisms in the checked simulator. Multiple compatible summaries would not identify one of them uniquely. No calibrated value is used here to claim that a fixed proportion of real state-weeks exceeds a biological or economic threshold.

5. Discussion

The theoretical result is a distinction between targets under an explicit observation model. With fixed, known reporting, exogenous exposure and conditionally independent counting errors, current-value risk is A/m and risk for the realised mean of observed periods is $A/(Tm)$. Estimating a stationary process mean additionally incurs the variance of that realised mean. This contribution is unchanged when cases increase within fixed periods, but its behaviour as periods accumulate depends on temporal covariance. Reporting volatility changes even the current and realised-mean risks. These identities provide a reference for interpreting case volume; their assumptions determine where that reference applies.

The variance structure has precedents in demographic and environmental stochasticity and in clustered sampling (Engen et al., 1998; Cochran, 1977). Its surveillance interpretation depends on what varies together and what the estimate is intended to describe. Independent offspring and independent reporting generate sampling error that decreases with exposure. Shared transmission conditions affect a process-mean estimate through the selection of periods. Shared reporting fluctuations affect the measurement of those periods. Calling every persistent component an environmental floor obscures this distinction and can mistakenly treat variation in the target as measurement error.

The conditional information calculation complements analyses of information in epidemic curves (Parag et al., 2022; Parag and Donnelly, 2022). It establishes efficiency of a scaled count in a specified unbiased estimator class with a known denominator and reporting law. It does not establish a universal lower bound for reproduction-number inference. Additional observations, independent reporting measurements and prior structure change the information available. Procedures that trade conditional bias for reduced variance also fall outside that class. A balance scale should therefore always be attached to its estimator, target and sampling experiment rather than interpreted as an intrinsic property of a pathogen.

A related distinction concerns estimation windows. In the Cori framework, widening a window accumulates infectiousness and incidence while imposing local constancy of the reproduction number over a longer interval (Cori et al., 2013; Gostic et al., 2020). This trades temporal resolution against precision and can introduce mismatch when transmission changes. It is not equivalent to drawing more independent transmission periods, nor does it preserve the estimand automatically. The repeated-period averages in Table 1 make a different assumption: each period has its own transmission value and a conditionally unbiased measurement. Applying their formulas to overlapping windows requires explicit treatment of overlap and temporal dependence.

The empirical analysis concerns squared relative deviations of apparent weekly statistics, not mean squared errors against observed R_t. The affine and power-law comparisons in Section 4.2 address the shape supported over the observed count range. A positive fitted intercept by itself does not establish a nonzero asymptote, and predictive similarity does not identify the mechanism generating either curve. Training-only centring and scoring prevent information leakage in predictive comparisons. Prospective curve-discrimination exercises remain conditional on their generators and address a different question from identifying biological transmission variance.

Controlled thinning adds a known sampling operation to recorded counts. It preserves the underlying observed trajectory before random removal, but nonlinear transformations, continuity corrections and random denominators can change both fitted coefficients. The derivative-based reference is an approximation to this pipeline, not an exact invariant of the statistic. Section 4.3 must therefore be interpreted alongside the retention-specific responses, uncertainty and known-count-model checks. Agreement with a reference response would support that local sampling description; it would not validate the full A/m identity for admissions or identify the origin of an unthinned slope. Failure to reject an intercept change is not evidence of equivalence.

State-intercept fits and alternative centring schemes answer separate questions. State intercepts allow baseline loss to differ across states while estimating an association with within-state count variation. They do not make that variation exogenous or remove common weekly shocks. Resampling summaries must match the intended state distribution, including sampled multiplicities; absent states have no estimated intercept. Recentring by season, trend or national week changes the loss being modelled. The resulting intercept changes are sensitivity measures, not orthogonal biological variance components. Any variance ratio for national-week means concerns signed deviations as defined in Methods, not automatically the squared-loss response.

Spatial and temporal dependence also constrain uncertainty. Regions observed in the same week share influences, while repeated observations within a region share history. Crossed resampling addresses these two sources of dependence under its own assumptions (Owen, 2007; Cameron et al., 2011); restricted time-block sensitivity explores residual serial dependence. Neither additional rows within the same season nor increasing admissions supplies independent seasons. Similarly, spatial aggregation can change the variability of transmission itself through covariance between local epidemics, as Eq. (7) shows. A cross-sectional count association can consequently combine sampling precision with differences in the populations being averaged.

Simulation-based inversion is conditional on the transmission and observation mechanism (Sections 3.6 and 4.5). The weekly reproduction-number variation specified in a daily renewal simulator is a model parameter indexed by calendar week. It is not directly observed per-generation truth, and multiplying case counts by a time conversion does not convert its variance or a balance scale. Independent observation dispersion, shared reporting shocks and transmission persistence must be specified separately. Agreement between statistics computed from the same admissions is a useful consistency check, but shared data and calibration assumptions limit its independence as evidence.

Generation intervals further condition this mapping. Serial intervals, intrinsic generation intervals and realised household generation intervals measure different quantities (Cowling et al., 2009; Vink et al., 2014; Chan et al., 2025). A sensitivity distribution with an assumed coefficient of variation must be distinguished from the source study's reported spread. Likewise, confidence limits for an interval mean do not define a posterior distribution for the entire generation interval. Conditional calibration intervals and changes across specified scenarios describe different uncertainties; a scenario range is not a joint confidence interval. Propagating uncertainty requires a justified distribution and an explicit joint model, not a relabelling of sensitivity endpoints.

Finally, facility participation measures only one aspect of ascertainment. Testing, care seeking, coding and admission practice can vary within participating hospitals. Comparisons across reporting regimes remain descriptive without a design identifying a policy effect. Independent measurements of these processes, finer spatial observations and further seasons would directly address the principal identification limits. The symbolic design extensions in Appendix Appendix B illustrate how a chosen target enters allocation, subject to costs, feasible periods and dependence. Operational choices require those quantities to be measured on consistent scales. The empirical summaries alone do not supply a universal case threshold or a uniquely identified transmission variance.

6. Conclusion

Case volume has different statistical value for different inferential targets. Under fixed, known reporting and exogenous exposure, counting error decreases for both the current reproduction number and the realised mean of observed periods. Estimating a stationary process mean additionally requires sampling periods whose transmission levels vary; dependence determines the remaining uncertainty. Unobserved reporting volatility contributes error even for realised targets, and count moments generally cannot separate it from transmission variation. Apparent weekly admission statistics measure deviations around a chosen centre rather than error against known transmission truth. Curve comparison, thinning, state-intercept fits and conditional calibration each address part of this observation problem, without identifying a universal case threshold. Surveillance design should specify the target, observation mechanism and temporal scale before using case volume to quantify precision or allocate resources.

Data and code availability

The NHSN Weekly Hospital Respiratory Data are publicly available from the U.S. Centers for Disease Control and Prevention (Centers for Disease Control and Prevention, 2026); the analyses used the snapshot retrieved on 2026-10-20. Analysis code, fixed analysis windows, simulation settings, versioned result bundles and the validation log are available at `https://github.com/wangchi2068/reproduction-number-targets-surveillance`. The release manifest records the SHA-256 of each raw input and generated asset. No individual-level or restricted data were used.

References

Auger-Méthé, M., Field, C., Albertsen, C.M., Derocher, A.E., Lewis, M.A., Jonsen, I.D., Mills Flemming, J., 2016. State-space models' dirty little secrets: even simple linear Gaussian models can have estimation problems. Scientific Reports 6, 26677. doi:10.1038/srep26677.

Azmon, A., Faes, C., Hens, N., 2014. On the estimation of the reproduction number based on misreported epidemic data. Statistics in Medicine 33, 1176–1192. doi:10.1002/sim.6015.

Bretó, C., He, D., Ionides, E.L., King, A.A., 2009. Time series analysis via mechanistic models. The Annals of Applied Statistics 3, 319–348. doi:10.1214/08-aoas201.

Cameron, A.C., Gelbach, J.B., Miller, D.L., 2011. Robust inference with multiway clustering. Journal of Business & Economic Statistics 29, 238–249. doi:10.1198/jbes.2010.07136.

Cauchemez, S., Valleron, A.J., Boëlle, P.Y., Flahault, A., Ferguson, N.M., 2008. Estimating the impact of school closure on influenza transmission from Sentinel data. Nature 452, 750–754. doi:10.1038/nature06732.

Centers for Disease Control and Prevention, 2026. Weekly hospital respiratory data (HRD) metrics by jurisdiction, national healthcare safety network (NHSN). URL: https://data.cdc.gov/Public-Health-Surveillance/Weekly-Hospital-Respiratory-Data-HRD-Metrics-by-Ju/ua7e-t2fy. data snapshot retrieved 2026-10-20.

Centers for Medicare & Medicaid Services, 2024. QSO-25-05-Hospitals/CAHs: Updates to the condition of participation (CoP) requirements for hospitals and critical access hospitals (CAHs) to report acute respiratory illnesses. URL: https://www.cms.gov/files/document/qso-25-05-hospitals-cahs.pdf. memorandum dated October 22, 2024.

Chan, L.Y.H., Morris, S.E., Stockwell, M.S., Bowman, N.M., Asturias, E., Rao, S., Lutrick, K., Ellingson, K.D., Nguyen, H.Q., Maldonado, Y., McLaren, S.H., Sano, E., Biddle, J.E., Smith-Jeffcoat, S.E., Biggerstaff, M., Rolfes, M.A., Talbot, H.K., Grijalva, C.G., Borchering, R.K., Mellis, A.M., RVTN-Sentinel Study Group, 2025. Estimating the generation time for influenza transmission using household data in the United States. Epidemics 50, 100815. doi:10.1016/j.epidem.2025.100815.

Cochran, W.G., 1977. Sampling Techniques. 3rd ed., John Wiley & Sons, New York.

Cori, A., Ferguson, N.M., Fraser, C., Cauchemez, S., 2013. A new framework and software to estimate time-varying reproduction numbers during epidemics. American Journal of Epidemiology 178, 1505–1512. doi:10.1093/aje/kwt133.

Cowling, B.J., Fang, V.J., Riley, S., Peiris, J.S.M., Leung, G.M., 2009. Estimation of the serial interval of influenza. Epidemiology 20, 344–347. doi:10.1097/ede.0b013e31819d1092.

Eales, O., Riley, S., 2024. Differences between the true reproduction number and the apparent reproduction number of an epidemic time series. Epidemics 46, 100742. doi:10.1016/j.epidem.2024.100742.

Engen, S., Bakke, Ø., Islam, A., 1998. Demographic and environmental stochasticity—concepts and definitions. Biometrics 54, 840–846. doi:10.2307/2533838.

Fraser, C., 2007. Estimating individual and household reproduction numbers in an emerging epidemic. PLoS ONE 2, e758. doi:10.1371/journal.pone.0000758.

Frutos, A.M., Moon, S., Binder, A.M., Cool, A.J., Iyawe, K., Thompson, T., Dudeck, M.A., Bozio, C.H., Tenforde, M.W., Biggerstaff, M., Sumner, K.M., 2026. Evaluation of a novel data source for national influenza surveillance: Influenza hospitalization data in the National Healthcare Safety Network, United States, September 2021–April 2024. American Journal of Public Health 116, 251–255. doi:10.2105/ajph.2025.308282.

Gostic, K.M., McGough, L., Baskerville, E.B., Abbott, S., Joshi, K., Tedijanto, C., Kahn, R., Niehus, R., Hay, J.A., De Salazar, P.M., Hellewell, J., Meakin, S., Munday, J.D., Bosse, N.I., Sherrat, K., Thompson, R.N., White, L.F., Huisman, J.S., Scire, J., Bonhoeffer, S., Stadler, T., Wallinga, J., Funk, S., Lipsitch, M., Cobey, S., 2020. Practical considerations for measuring the effective reproductive number, R_t. PLOS Computational Biology 16, e1008409. doi:10.1371/journal.pcbi.1008409.

Hamilton, J.D., 1994. Time Series Analysis. Princeton University Press, Princeton, NJ.

Hayes, R.J., Bennett, S., 1999. Simple sample size calculation for cluster-randomized trials. International Journal of Epidemiology 28, 319–326. doi:10.1093/ije/28.2.319.

He, D., Ionides, E.L., King, A.A., 2010. Plug-and-play inference for disease dynamics: measles in large and small populations as a case study. Journal of the Royal Society Interface 7, 271–283. doi:10.1098/rsif.2009.0151.

Knape, J., 2008. Estimability of density dependence in models of time series data. Ecology 89, 2994–3000. doi:10.1890/08-0071.1.

Lande, R., Engen, S., Sæther, B.E., 2003. Stochastic Population Dynamics in Ecology and Conservation. Oxford University Press, Oxford.

Lehmann, E.L., Casella, G., 1998. Theory of Point Estimation. 2nd ed., Springer, New York.

Lloyd-Smith, J.O., Schreiber, S.J., Kopp, P.E., Getz, W.M., 2005. Superspreading and the effect of individual variation on disease emergence. Nature 438, 355–359. doi:10.1038/nature04153.

Nash, R.K., Bhatt, S., Cori, A., Nouvellet, P., 2023. Estimating the epidemic reproduction number from temporally aggregated incidence data: A statistical modelling approach and software tool. PLOS Computational Biology 19, e1011439. doi:10.1371/journal.pcbi.1011439.

Ogi-Gittins, I., Hart, W.S., Song, J., Nash, R.K., Polonsky, J., Cori, A., Hill, E.M., Thompson, R.N., 2024. A simulation-based approach for estimating the time-dependent reproduction number from temporally aggregated disease incidence time series data. Epidemics 47, 100773. doi:10.1016/j.epidem.2024.100773.

Owen, A.B., 2007. The pigeonhole bootstrap. The Annals of Applied Statistics 1, 386–411. doi:10.1214/07-aoas122.

Parag, K.V., 2021. Improved estimation of time-varying reproduction numbers at low case incidence and between epidemic waves. PLOS Computational Biology 17, e1009347. doi:10.1371/journal.pcbi.1009347.

Parag, K.V., Donnelly, C.A., 2022. Fundamental limits on inferring epidemic resurgence in real time using effective reproduction numbers. PLOS Computational Biology 18, e1010004. doi:10.1371/journal.pcbi.1010004.

Parag, K.V., Donnelly, C.A., Zarebski, A.E., 2022. Quantifying the information in noisy epidemic curves. Nature Computational Science 2, 584–594. doi:10.1038/s43588-022-00313-1.

Parag, K.V., Lambert, B., Donnelly, C.A., Beregi, S., 2025. Asymmetric limits on timely interventions from noisy epidemic data. Communications Physics 8, 450. doi:10.1038/s42005-025-02358-w.

Raudenbush, S.W., 1997. Statistical analysis and optimal design for cluster randomized trials. Psychological Methods 2, 173–185. doi:10.1037/1082-989x.2.2.173.

Searle, S.R., Casella, G., McCulloch, C.E., 1992. Variance Components. John Wiley & Sons, New York.

Shaman, J., Kohn, M., 2009. Absolute humidity modulates influenza survival, transmission, and seasonality. Proceedings of the National Academy of Sciences 106, 3243–3248. doi:10.1073/pnas.0806852106.

Stoner, O., Economou, T., Drummond Marques da Silva, G., 2019. A hierarchical framework for correcting underreporting in count data. Journal of the American Statistical Association 114, 1481–1492. doi:10.1080/01621459.2019.1573732.

Thompson, R.N., Stockwin, J.E., van Gaalen, R.D., Polonsky, J.A., Kamvar, Z.N., Demarsh, P.A., Dahlqwist, E., Li, S., Miguel, E., Jombart, T., Lessler, J., Cauchemez, S., Cori, A., 2019. Improved inference of time-varying reproduction numbers during infectious disease outbreaks. Epidemics 29, 100356. doi:10.1016/j.epidem.2019.100356.

Vink, M.A., Bootsma, M.C.J., Wallinga, J., 2014. Serial intervals of respiratory infectious diseases: A systematic review and analysis. American Journal of Epidemiology 180, 865–875. doi:10.1093/aje/kwu209.

Wallinga, J., Lipsitch, M., 2007. How generation intervals shape the relationship between growth rates and reproductive numbers. Proceedings of the Royal Society B: Biological Sciences 274, 599–604. doi:10.1098/rspb.2006.3754.

Appendix A. Proofs

Appendix A.1. Count law and conditional information

The probability generating function of one negative binomial offspring count is $[1 + (R_t/k)(1 - z)]^{-k}$. Conditional independence raises this function to the power n. Substituting $z \mapsto 1 - \rho + \rho z$ for independent reporting gives $[1 + (\rho R_t/k)(1 - z)]^{-nk}$, proving Eq. (2). Its derivatives give the stated moments.

For fixed, known (n, ρ, k) and $R > 0$, the count likelihood has score and Fisher information

$$\dot{\ell}(R) = \frac{c}{R} - \frac{(c + nk)\rho}{k + \rho R} = \mathcal{I}(R)(\widehat{R} - R), \qquad \mathcal{I}(R) = \frac{m}{R + \rho R^2/k}. \tag{Appendix A.1}$$

Consequently $\mathrm{Var}(\widehat{R} \mid R) = \mathcal{I}(R)^{-1}$. The support does not depend on R, and the regular one-parameter Cramér–Rao inequality applies to estimators unbiased for every R. It gives the attained bound in Section 2 (Lehmann and Casella, 1998). The calculation does not establish an unrestricted decision-theoretic bound or a bound with unknown nuisance parameters.

Appendix A.2. Proof of Proposition 1

Let $e_t = \widehat{R}_t - R_t$. Then $\mathbb{E}(e_t \mid R_t) = 0$ and

$$\mathbb{E}(e_t^2) = \frac{\mathbb{E}R_t + \rho\mathbb{E}R_t^2/k}{m} = \frac{A}{m}, \qquad \mathbb{E}[e_t(R_t - \bar{R})] = 0. \tag{Appendix A.2}$$

Squaring Eq. (4) proves Eq. (3). For any estimator based on the same count and unbiased conditional on every R_t, its expected conditional variance is at least $\mathbb{E}[\mathcal{I}(R_t)^{-1}] = A/m$. Its process-mean risk adds the same v_R, proving the restricted optimality statement. At fixed ρ, k and transmission law,

$$\frac{V(m) - V(2m)}{V(m)} = \frac{1}{2(1 + m/m_\times)}, \qquad v_R > 0. \tag{Appendix A.3}$$

This comparison changes exposure n; it does not hold unchanged when reporting probability changes.

Appendix A.3. Proof of Corollary 1

Conditional on the entire path $R_{1:T}$, the e_t are independent and centred. They are therefore pairwise uncorrelated and orthogonal to every square-integrable function of that path. Thus

$$\mathbb{E}\left[\left(T^{-1}\sum_t e_t\right)^2\right] = \frac{A}{Tm}, \qquad \mathrm{Var}(\widehat{\bar{R}}_T) = \mathrm{Var}(\bar{R}_T) + \frac{A}{Tm}. \tag{Appendix A.4}$$

For independent, identically distributed R_t, $\mathrm{Var}(\bar{R}_T) = v_R/T$. Under the stated stationary autocovariance, summing all covariance pairs gives $T^2\mathrm{Var}(\bar{R}_T) = v_R[T + 2\sum_{h=1}^{T-1}(T - h)\phi^h]$. The finite geometric sum yields Eq. (6), including $T = 1$ and $\phi = 0$ (Hamilton, 1994). Similarly, summing spatial covariance pairs gives Eq. (7); its correlation matrix must be admissible.

Appendix A.4. Proof of Proposition 2 and Corollary 2

Write $z_t = p_t/\rho$ and $u_t = C_t/m - z_t R_t$. Conditional on (R_t, p_t), u_t has zero mean. Independence of p_t and R_t, with $\mathbb{E}z_t = 1$ and $\mathbb{E}z_t^2 = 1 + \tau^2$, gives

$$\mathbb{E}(u_t^2) = \frac{\bar{R} + \rho(1 + \tau^2)(\bar{R}^2 + v_R)/k}{m} = \frac{A_H}{m}. \tag{Appendix A.5}$$

The error $\widehat{R}_t - R_t = u_t + (z_t - 1)R_t$ has uncorrelated terms. The second has expected square $\tau^2(\bar{R}^2 + v_R) = v_H$, proving Eq. (9). Moreover $\mathbb{E}(\widehat{R}_t - R_t \mid R_t) = 0$, so process-mean risk adds v_R. Under the additional cross-period assumptions of Corollary 2, these errors are centred and independent conditional on the transmission path after integrating out reporting, giving Eq. (10).

For the identification claim, set $U = (1 + \tau^2)(\bar{R}^2 + v_R)$. The two observable moments of the scaled count satisfy

$$\mathbb{E}\widehat{R}_t = \bar{R}, \qquad \mathrm{Var}(\widehat{R}_t) = U - \bar{R}^2 + \frac{\bar{R} + \rho U/k}{m}. \tag{Appendix A.6}$$

Whenever admissible distinct values of τ^2 give $v_R = U/(1 + \tau^2) - \bar{R}^2 \geq 0$, these moments coincide. Such distinct pairs occur away from degenerate boundaries; varying exogenous m does not separate them. The restrictions on p_t and R_t must still hold. This proves nonidentification by these moments, without asserting equality of full count distributions.

For $v_R > 0$, the generalised balance scale $m_{\times,H} = A_H/(v_R + v_H)$ obeys

$$m_\times - m_{\times,H} = \frac{\tau^2(\bar{R}^2 + v_R)\bar{R}(1 + \rho\bar{R}/k)}{v_R(v_R + v_H)} \geq 0. \tag{Appendix A.7}$$

For $\bar{R} > 0$, equality holds exactly when $\tau = 0$. A lower balance scale here accompanies a larger risk, not more accurate surveillance.

Appendix B. Symbolic theoretical extensions

Appendix B.1. Forecasting and shrinkage

Under fixed, known reporting and the stationary covariance model of Corollary 1, assume the counting error at t is centred conditional on the entire transmission path. Then

$$\mathbb{E}[(\widehat{R}_t - R_{t+1})^2] = \frac{A}{m} + 2(1 - \phi)v_R. \tag{Appendix B.1}$$

For known $\bar{R}$, v_R and ϕ, minimising over forecasts of the form $\bar{R} + \lambda(\widehat{R}_t - \bar{R})$ gives

$$\lambda^* = \frac{\phi v_R}{v_R + A/m}, \qquad \text{MSE}_{\text{lin}} = v_R - \frac{\phi^2 v_R^2}{v_R + A/m}. \tag{Appendix B.2}$$

These follow by expanding a quadratic in λ and using $\text{Cov}(\widehat{R}_t, R_{t+1}) = \phi v_R$. The limit is $(1 - \phi^2)v_R$; with independent transmission periods, the best forecast in this class is the known process mean. Optimality is restricted to linear forecasts from the single current estimate, not all histories or nonlinear predictors.

For estimating $\bar{R}$ instead, consider $\hat{\theta}_\lambda = (1 - \lambda)\widehat{R}_t + \lambda\mu_0$ with fixed benchmark μ_0 and $\Delta = \mu_0 - \bar{R}$. Its risk is

$$\mathbb{E}[(\hat{\theta}_\lambda - \bar{R})^2] = (1 - \lambda)^2 V + \lambda^2\Delta^2, \qquad V = v_R + A/m. \tag{Appendix B.3}$$

The oracle weight $\lambda^* = V/(V + \Delta^2)$ gives risk $V\Delta^2/(V + \Delta^2)$. This may be below v_R, illustrating the importance of the unbiasedness restriction. The oracle uses unknown quantities; estimating its weight requires a separate risk analysis.

Appendix B.2. Budget allocation with feasible periods

Proposition 3 (Continuous budget relaxation)**.** *Under H1–H4, independent, identically distributed transmission periods, conditionally independent counts, fixed ρ and linear costs, let $\mathcal{B} = T(c_0 + c_1 m)$ with $c_0, c_1 > 0$. Treat $T \geq 1$ and $m > 0$ as continuous. If $v_R > 0$ and $\mathcal{B} > c_0$, the unconstrained stationary point is*

$$m^* = \sqrt{\frac{c_0 A}{c_1 v_R}} = \sqrt{\frac{c_0}{c_1} m_\times}. \tag{Appendix B.4}$$

The feasible optimum is $\min\{m^*, (\mathcal{B} - c_0)/c_1\}$*, with T recovered from the budget constraint. A prescribed lower bound $m_{\min} > 0$ instead requires clamping m^* to $[m_{\min}, (\mathcal{B} - c_0)/c_1]$, provided this interval is nonempty.*

Proof. For independent, identically distributed periods, process-mean risk is

$$\frac{v_R}{T} + \frac{A}{Tm} = \frac{(c_0 + c_1 m)(v_R + A/m)}{\mathcal{B}} =: f(m). \tag{Appendix B.5}$$

Its derivatives are $f'(m) = (c_1 v_R - c_0 A/m^2)/\mathcal{B}$ and $f''(m) = 2c_0 A/(\mathcal{B}m^3) > 0$. Strict convexity gives the stationary point and the clamped solution. When $v_R = 0$, risk decreases with m, so the optimum is the largest feasible m, corresponding to $T = 1$ without further constraints. □

The relaxation has the familiar form of cluster-size allocation (Cochran, 1977; Raudenbush, 1997). Actual designs require integer T and integer n, with $m = \rho n$. Under a budget inequality, for each feasible integer T choose

$$n_T = \left\lfloor \frac{\mathcal{B}/T - c_0}{c_1\rho} \right\rfloor \geq 1, \qquad m_T = \rho n_T, \tag{Appendix B.6}$$

and compare the risks over all feasible T. This handles unspent budget and the boundary $T = 1$; rounding the relaxed optimum alone need not suffice.

Appendix B.3. Budget allocation under temporal dependence

For the stationary autocovariance model, compare instead the exact integer-design objective

$$F(T) = \frac{v_R S(\phi, T)}{T^2} + \frac{A}{T m_T}. \tag{Appendix B.7}$$

No independent-period optimum or universal switch threshold follows from this expression. Under independent reporting shocks, replace its counting term by $(v_H + A_H/m_T)/T$. All costs and exposures must refer to the same period; these symbolic formulas provide no direct conversion of weekly admission ratios into generational design thresholds.

Appendix B.4. Coverage and reporting completeness

For fixed transmission law and a differentiable reporting-volatility relation $\tau^2(\rho)$ independent of n, write $Q = \bar{R}^2 + v_R$. At interior values of ρ, the reductions in single-period process-mean risk from proportional increases in n and ρ are

$$\begin{aligned} -\frac{\partial V_H}{\partial \log n} &= \frac{\bar{R}}{\rho n} + \frac{(1+\tau^2)Q}{kn}, \\ -\frac{\partial V_H}{\partial \log \rho} &= \frac{\bar{R}}{\rho n} - \rho \frac{\mathrm{d}\tau^2}{\mathrm{d}\rho} Q \left(1 + \frac{1}{kn}\right). \end{aligned} \tag{Appendix B.8}$$

These follow by differentiating Eq. (8) before substituting $m = \rho n$. At fixed τ, the first gain is larger. If improving completeness also stabilises reporting, that ordering can change; these are proportional, not cost-adjusted, comparisons.

Corollary 3 (A specified reporting mechanism)**.** *If p_t is the fraction of ν independent reporting units participating with probability ρ, so $\tau^2 = (1-\rho)/(\rho\nu)$, increasing completeness has the larger proportional gain exactly when*

$$m > \frac{\rho^2(\nu - 1)}{k}. \tag{Appendix B.9}$$

For $p_t \sim \mathrm{Beta}(\rho\nu, (1-\rho)\nu)$, the corresponding condition is $m > \rho^2\nu/k$.

Subtracting the first derivative gain from the second gives $Q(kn)^{-1}[(kn+\rho)/(\rho\nu) - 1]$ for the reporting-unit model. Replacing ν by $\nu + 1$ proves the beta case. Neither mechanism is implied by observed facility participation alone.

Appendix B.5. Conditional renewal moments and noisy denominators

A restricted renewal analogue can be stated explicitly. Given history $\mathcal{F}$, suppose infection pressure $\Lambda > 0$ is independent of the next R, with $\mathbb{E}(R \mid \mathcal{F}) = \bar{R}$ and $\mathrm{Var}(R \mid \mathcal{F}) = v_R$. Assume a conditional count law with mean $\rho\Lambda R$ and variance $\rho\Lambda R + \rho^2\Lambda R^2/k$. Put $\bar{\Lambda} = \mathbb{E}(\Lambda \mid \mathcal{F})$ and $m = \rho\bar{\Lambda}$. Total variance gives

$$\frac{\mathrm{Var}(C \mid \mathcal{F})}{m^2} = v_R + \frac{A}{m} + \frac{\mathrm{Var}(\Lambda \mid \mathcal{F})}{\bar{\Lambda}^2}(\bar{R}^2 + v_R). \tag{Appendix B.10}$$

The result assumes the conditional independence and fixed moments just stated. It is a count-prediction variance, not a general error identity for a ratio with an estimated denominator, and it does not resolve endogenous incidence.

To see the ratio issue, condition on a positive latent path and write numerator and denominator as their latent values times $(1+\varepsilon)$ and $(1+u)$. Let $Y = y(1+\varepsilon)/(1+u)$, with centred disturbances of order τ, a denominator bounded away from zero, bounded scaled third moments and integrable Taylor remainders. Then

$$\begin{aligned} \mathbb{E}(Y \mid y, \text{path})/y &= 1 + \mathbb{E}(u^2 \mid \text{path}) - \mathbb{E}(\varepsilon u \mid \text{path}) + O(\tau^3), \\ \mathrm{Var}(Y \mid \text{path})/y^2 &= \mathrm{Var}(\varepsilon - u \mid \text{path}) + O(\tau^3). \end{aligned} \tag{Appendix B.11}$$

Both numerator–denominator covariance and induced bias enter apparent variation. Overlapping ratios can share disturbances. These conditional approximations do not supply an exact source decomposition of a fitted loss intercept.

Appendix B.6. Replication and variance estimation

Proposition 4 (A conditional information bound for process variance)**.** *Suppose $R_1, \ldots, R_T$ are independent, identically distributed gamma variables with known mean $\bar{R}$ and variance $v_R > 0$. If the observation law given $R_{1:T}$ does not otherwise depend on v_R, any regular unbiased estimator of v_R from the counts has relative standard error at least*

$$[T\xi(\xi\psi'(\xi) - 1)]^{-1/2}, \qquad \xi = \bar{R}^2/v_R, \tag{Appendix B.12}$$

where ψ' is the trigamma function.

Observing the latent R_t cannot provide less information than observing only counts generated from them. With the mean fixed, information for gamma shape ξ is $\psi'(\xi) - 1/\xi$ per period; the change of parameter $\mathrm{d}\xi/\mathrm{d}v_R = -\xi/v_R$ gives the bound. It approaches $\sqrt{2/T}$ as $\xi \to \infty$.

Corollary 4 (Independent season effects)**.** *For $S \geq 2$ independent, identically distributed Gaussian season effects with unknown mean and variance $v_S > 0$, assume their conditional count law is parameter-free. Any estimator unbiased for v_S for all means and variances has relative standard error at least $\sqrt{2/(S-1)}$.*

Conditional expectation onto the latent effects cannot increase variance. Their sample variance is uniformly minimum-variance unbiased and has variance $2v_S^2/(S-1)$ (Lehmann and Casella, 1998; Searle et al., 1992). Thus more observations within existing seasons cannot replace independent seasons for this bound. Both bounds are conditional model results, not empirical confidence limits.

Appendix C. Supplementary methods and results

Appendix C.1. Input registry and reproducibility

All reported analyses are regenerated from the raw NHSN state-week table whose SHA-256 is recorded in the release manifest. The panel reconstructs missing values as missing and requires four consecutive weekly dates; it does not insert zero admissions. The primary samples and all fixed windows are identified by deterministic keys in the release. Every retained table, figure and inline macro is generated from the same versioned result bundle; the manifest records input hashes, generator hash, actual bootstrap budgets, fit failures and Monte Carlo diagnostics. A resumed computation must match the complete source, input and configuration fingerprint; otherwise it is run afresh.

Table S1 reports pooled affine working fits and the three generation-interval transforms. The 3.6-day and fixed-coefficient-of-variation variants are stated as choices; the 3.2-day, SD 2.1-day variant reproduces the intrinsic interval estimate of Chan et al. (2025). Table S2 reports fold-level held-out scores from training-only centres and fits. The season folds use the same data as the fit and are descriptive.

The calibration registry declares each input as a documented external estimate, a value derived from the observed panel, or an analysis assumption. Delay, offspring dispersion, admission probability and the weekly shock variance are assumptions in the simulator; their structural role is sensitivity, not measurement. In the observation model, binomial admission allocation is the baseline and $\varphi > 1$ describes direct gamma–Poisson counts with conditional variance $\varphi\lambda$. The simulator is checked for conditional observation moments and for the direct-count construction; those tests verify implementation, not empirical adequacy.

Appendix C.2. Conditional calibration

Each declared scenario is simulated on the four fixed observed window lengths with the observed first-week state counts. Initial zero counts remain zero and are not raised; this can reduce the retained sample relative to the empirical panel, and the discrepancy is retained in the response diagnostics. Weekly reproduction shocks separate national and state components; persistence in reporting is a separate mechanism. Reproduction-number innovations are truncated below at a declared positive bound, and the truncation fraction is recorded.

For each empirical statistic, a weighted line relates the simulated mean intercept to the declared weekly variance grid. Paired empirical draws and paired simulated panels are combined to obtain conditional inversion draws. Failed fits, nonfinite values, nonpositive implied values and grid extrapolation are counted and reported; percentile intervals refer only to finite draws and are labelled conditional. Several scenarios can give similar empirical summaries while differing in observation mechanism. The calibration therefore reports response checks for intercept, slope, sample size, signed-deviation variance ratio and within-state and national lag-one autocorrelation. It does not assign a definitive biological variance or a probability that one mechanism is correct.

Appendix C.3. Secondary sensitivities and design illustration

Table S8 reports season-specific intercepts and a paired participation adjustment in which each statistic's counts are divided by facility participation while the original scale is retained. The adjustment measures only the recorded participation proxy; testing intensity, coding and care seeking inside reporting facilities are not observed. The three summer summaries are unadjusted and describe a single voluntary period; they are not causal comparisons or independent replications. No equivalence or mandate effect is claimed.

Table S7 reports a model-conditional illustration of curve discrimination. Observed season designs are recycled with declared count multipliers and crossed mean-one lognormal variation; the generating curves are the fitted affine and power-law means. Calibration quantiles and evaluation simulations are independent, and the null rejection fraction is reported with an exact binomial interval before power is interpreted. Power was near one in every cell. That saturation shows the synthetic generator is far more favourable than the observed data, where the two curves were not distinguishable; the exercise therefore does not answer how many seasons would be needed in practice and is reported only to document that limitation. The budget expressions in Appendix Appendix B.2 remain symbolic; no weekly admission ratio is converted into a generational design threshold.

Table S1. Pooled affine loss fits and generation-interval sensitivities

Statistic	Parameter	n	Estimate [95% CI]	Finite/B
RA	a	1748	0.01495 [0.01001, 0.02092]	2000/2000
RA	b	1748	0.56 [0.369, 0.7692]	2000/2000
RA	b/a	1748	37.44 [18.93, 69.92]	2000/2000
G	a	1756	0.08401 [0.04986, 0.1234]	2000/2000
G	b	1756	1.971 [0.9889, 3.042]	2000/2000
G	b/a	1756	23.46 [8.279, 54.7]	2000/2000
RA32	a	1748	0.01191 [0.007997, 0.01668]	2000/2000
RA32	b	1748	0.4482 [0.2975, 0.6147]	2000/2000
RA32	b/a	1748	37.62 [19.13, 69.89]	2000/2000
RA32source	a	1748	0.01106 [0.007501, 0.01553]	2000/2000
RA32source	b	1748	0.4196 [0.2827, 0.5688]	2000/2000
RA32source	b/a	1748	37.93 [19.42, 69.26]	2000/2000

Signed percentile intervals; ratios are defined only for positive denominators. Finite-draw counts are retained for every estimate. RA a: 2000/2000 finite draws; RA b: 2000/2000 finite draws; RA ratio: 2000/2000 finite draws; G a: 2000/2000 finite draws; G b: 2000/2000 finite draws; G ratio: 2000/2000 finite draws; RA32 a: 2000/2000 finite draws; RA32 b: 2000/2000 finite draws; RA32 ratio: 2000/2000 finite draws; RA32source a: 2000/2000 finite draws; RA32source b: 2000/2000 finite draws; RA32source ratio: 2000/2000 finite draws.

Table S2. Foldwise model validation

Statistic / split	Fold	Valid/total or n	Affine	Power	Difference	Wins / valid
RA / state	Summary	51/51	–	–	-0.002018	26
RA / season	Summary	4/4	–	–	0.004626	3
RA	2022-23	365	-2.378	-2.369	0.008954	yes
RA	2023-24	499	-2.613	-2.612	0.0008806	yes
RA	2024-25	356	-2.46	-2.481	-0.02077	yes
RA	2025-26	528	-2.346	-2.324	0.0223	yes
G / state	Summary	51/51	–	–	-0.007369	22
G / season	Summary	4/4	–	–	-0.004414	2
G	2022-23	367	-0.5979	-0.5942	0.003756	yes
G	2023-24	503	-1.201	-1.192	0.008487	yes
G	2024-25	357	-0.7333	-0.7584	-0.02505	yes
G	2025-26	529	-0.7557	-0.7641	-0.008421	yes
RA32 / state	Summary	51/51	–	–	-0.002242	25
RA32 / season	Summary	4/4	–	–	0.004343	3
RA32	2022-23	365	-2.612	-2.604	0.00805	yes
RA32	2023-24	499	-2.83	-2.83	0.0001619	yes
RA32	2024-25	356	-2.691	-2.712	-0.02118	yes
RA32	2025-26	528	-2.572	-2.549	0.02294	yes
RA32source / state	Summary	51/51	–	–	-0.002824	25
RA32source / season	Summary	4/4	–	–	0.003577	2
RA32source	2022-23	365	-2.701	-2.695	0.005641	yes
RA32source	2023-24	499	-2.886	-2.888	-0.001735	yes
RA32source	2024-25	356	-2.77	-2.792	-0.02217	yes
RA32source	2025-26	528	-2.644	-2.619	0.02453	yes

Summary rows show valid/total folds and the number favoring affine in the last column; season rows show test n and convergence. Score is mean L/mu + log(mu), lower is better. Difference = power minus affine, so positive favors affine. The summary difference is the mean across folds weighted by held-out test n; the unweighted fold mean and the count of folds favouring affine are shown separately. Fold fits refit the training center. Training sets overlap; no iid-fold SE or four-season significance test is reported.

Table S3. Shared-count binomial thinning of the R_A sample (n=1748): estimates, simultaneous bands and simulation diagnostics

Sample	q / fit	Metric	Estimate [95% CI]	Simultaneous 95%	Finite/B
full	q=0.8	b	0.6747 [0.4905, 0.9034]	–	1000/1000
full	q=0.8	c	0.4588 [0.3731, 0.5557]	[0.3381, 0.5795]	1000/1000
full	q=0.8	delta_a	-0.0003183 [-0.000786, 0.00005128]	[-0.0008833, 0.0002467]	1000/1000
full	q=0.8	c_over_bP	1.03 [0.8476, 1.242]	–	1000/1000
full	q=0.6	b	0.8763 [0.6941, 1.103]	–	1000/1000
full	q=0.6	c	0.4745 [0.4064, 0.5532]	[0.3751, 0.574]	1000/1000
full	q=0.6	delta_a	-0.000836 [-0.001963, -0.000006555]	[-0.00213, 0.0004583]	1000/1000
full	q=0.6	c_over_bP	1.065 [0.915, 1.241]	–	1000/1000
full	q=0.5	b	1.057 [0.858, 1.275]	–	1000/1000
full	q=0.5	c	0.4969 [0.4243, 0.5606]	[0.4047, 0.589]	1000/1000
full	q=0.5	delta_a	-0.001318 [-0.002744, -0.00009326]	[-0.003126, 0.0004903]	1000/1000
full	q=0.5	c_over_bP	1.115 [0.9621, 1.252]	–	1000/1000
full	q=0.4	b	1.31 [1.112, 1.534]	–	1000/1000
full	q=0.4	c	0.4997 [0.4397, 0.5611]	[0.4158, 0.5837]	1000/1000
full	q=0.4	delta_a	-0.001794 [-0.00376, -0.0001502]	[-0.004272, 0.0006837]	1000/1000
full	q=0.4	c_over_bP	1.122 [0.9903, 1.257]	–	1000/1000
full	q=0.3	b	1.742 [1.503, 1.958]	–	1000/1000
full	q=0.3	c	0.5067 [0.4482, 0.5539]	[0.4326, 0.5808]	1000/1000
full	q=0.3	delta_a	-0.002454 [-0.004852, -0.000159]	[-0.005728, 0.0008198]	1000/1000
full	q=0.3	c_over_bP	1.137 [1.008, 1.249]	–	1000/1000
full	q=0.2	b	2.46 [2.221, 2.711]	–	1000/1000
full	q=0.2	c	0.4749 [0.4292, 0.5229]	[0.4126, 0.5373]	1000/1000
full	q=0.2	delta_a	-0.002484 [-0.005751, 0.0005387]	[-0.006867, 0.001899]	1000/1000
full	q=0.2	c_over_bP	1.066 [0.9728, 1.173]	–	1000/1000
full	q=0.1	b	3.983 [3.509, 4.379]	–	1000/1000
full	q=0.1	c	0.3804 [0.3337, 0.4204]	[0.3234, 0.4373]	1000/1000
full	q=0.1	delta_a	0.0007312 [-0.00305, 0.005984]	[-0.005595, 0.007057]	1000/1000
full	q=0.1	c_over_bP	0.8537 [0.7415, 0.9566]	–	1000/1000
full	full	c	0.4051 [0.3656, 0.4413]	–	1000/1000
full	full	c_over_bP	0.9093 [0.8101, 0.9995]	–	1000/1000
full	full	delta_a_rate	-0.0001293 [-0.0006192, 0.0004699]	–	1000/1000
full	noq01	c	0.4848 [0.4387, 0.5279]	–	1000/1000
full	noq01	c_over_bP	1.088 [0.9936, 1.186]	–	1000/1000
full	noq01	delta_a_rate	-0.0008058 [-0.001703, 0.00005179]	–	1000/1000
highCt50	q=0.8	b	0.7797 [-0.007567, 1.573]	–	1000/1000
highCt50	q=0.8	c	0.442 [0.2771, 0.6691]	[0.1838, 0.7002]	1000/1000
highCt50	q=0.8	delta_a	-0.00007119 [-0.0004223, 0.0001553]	[-0.0004703, 0.0003279]	1000/1000
highCt50	q=0.8	c_over_bP	1.051 [0.6634, 1.606]	–	1000/1000
highCt50	q=0.6	b	0.969 [0.1971, 1.773]	–	1000/1000
highCt50	q=0.6	c	0.4497 [0.3339, 0.6318]	[0.256, 0.6433]	1000/1000
highCt50	q=0.6	delta_a	-0.0001877 [-0.0009664, 0.0002652]	[-0.0009926, 0.0006171]	1000/1000
highCt50	q=0.6	c_over_bP	1.069 [0.8015, 1.51]	–	1000/1000
highCt50	q=0.5	b	1.126 [0.3593, 1.891]	–	1000/1000
highCt50	q=0.5	c	0.4567 [0.3405, 0.6213]	[0.273, 0.6404]	1000/1000
highCt50	q=0.5	delta_a	-0.0002913 [-0.001459, 0.0003529]	[-0.001456, 0.000873]	1000/1000
highCt50	q=0.5	c_over_bP	1.086 [0.8187, 1.476]	–	1000/1000
highCt50	q=0.4	b	1.351 [0.6322, 2.134]	–	1000/1000
highCt50	q=0.4	c	0.4548 [0.3568, 0.593]	[0.2972, 0.6124]	1000/1000
highCt50	q=0.4	delta_a	-0.0004078 [-0.001854, 0.0004131]	[-0.001938, 0.001122]	1000/1000
highCt50	q=0.4	c_over_bP	1.081 [0.8483, 1.413]	–	1000/1000
highCt50	q=0.3	b	1.723 [1.005, 2.495]	–	1000/1000
highCt50	q=0.3	c	0.4515 [0.3668, 0.5711]	[0.3176, 0.5854]	1000/1000
highCt50	q=0.3	delta_a	-0.0005649 [-0.00247, 0.0006027]	[-0.002578, 0.001448]	1000/1000
highCt50	q=0.3	c_over_bP	1.073 [0.8687, 1.355]	–	1000/1000
highCt50	q=0.2	b	2.522 [1.77, 3.276]	–	1000/1000
highCt50	q=0.2	c	0.4633 [0.3813, 0.5611]	[0.3435, 0.5831]	1000/1000
highCt50	q=0.2	delta_a	-0.0009427 [-0.00351, 0.0008025]	[-0.003822, 0.001937]	1000/1000
highCt50	q=0.2	c_over_bP	1.102 [0.9074, 1.325]	–	1000/1000
highCt50	q=0.1	b	4.97 [4.216, 5.814]	–	999/1000
highCt50	q=0.1	c	0.4778 [0.4127, 0.5552]	[0.3835, 0.5721]	999/1000
highCt50	q=0.1	delta_a	-0.001594 [-0.005615, 0.001213]	[-0.006149, 0.002962]	999/1000
highCt50	q=0.1	c_over_bP	1.136 [0.9843, 1.323]	–	999/1000
highCt50	full	c	0.4734 [0.409, 0.5503]	–	999/1000
highCt50	full	c_over_bP	1.126 [0.9702, 1.316]	–	999/1000
highCt50	full	delta_a_rate	-0.0001928 [-0.0006891, 0.0001441]	–	999/1000
highCt50	noq01	c	0.4594 [0.3808, 0.5603]	–	1000/1000
highCt50	noq01	c_over_bP	1.092 [0.9026, 1.327]	–	1000/1000
highCt50	noq01	delta_a_rate	-0.0002434 [-0.0009563, 0.0002083]	–	1000/1000
both50	q=0.8	b	0.5473 [-0.2021, 1.391]	–	1000/1000
both50	q=0.8	c	0.4994 [0.3073, 0.7827]	[0.1979, 0.8009]	1000/1000

Table S4. Fixed effects for the R_A sample (n=1748): paired ratios, block sensitivity and convergence

Block	Parameter	Estimate [95% CI]	Finite/B	Nonpositive / diagnostic
L=1	Pooled a	0.01495 [0.009976, 0.02114]	1000/1000	0
L=1	Pooled b	0.56 [0.3682, 0.7707]	1000/1000	0
L=1	Pooled b/a	37.44 [18.64, 73.92]	1000/1000	0
L=1	State FE b	0.3794 [0.1082, 0.6403]	999/1000	0
L=1	State FE mean a	0.02192 [0.01289, 0.03502]	999/1000	0
L=1	State FE b/mean a	17.3 [3.315, 47.8]	999/1000	0
L=1	State FE median a	0.01751 [0.01037, 0.03014]	999/1000	0
L=1	State FE b/median a	21.67 [3.977, 57.85]	999/1000	0
L=1	State FE row-weighted a	0.02061 [0.01339, 0.03114]	999/1000	0
L=1	State FE row-weighted ratio	18.41 [3.705, 45.56]	999/1000	0
L=1	State FE count-weighted a	0.01542 [0.01056, 0.02288]	999/1000	0
L=1	State FE count-weighted ratio	24.61 [5.265, 55.65]	999/1000	0
L=1	True D_t b	0.3691 [0.1153, 0.6145]	999/1000	0
L=1	True D_t mean a	0.02084 [0.01218, 0.03429]	999/1000	0
L=1	True D_t ratio	17.71 [3.267, 47.81]	999/1000	0
L=1	Paired pooled/FE ratio	2.164 [1.04, 7.745]	999/1000	0
L=2	Pooled a	0.01495 [0.009493, 0.02077]	1000/1000	0
L=2	Pooled b	0.56 [0.371, 0.763]	1000/1000	0
L=2	Pooled b/a	37.44 [18.73, 74.77]	1000/1000	0
L=2	State FE b	0.3794 [0.109, 0.6611]	1000/1000	0
L=2	State FE mean a	0.02192 [0.01288, 0.03581]	1000/1000	0
L=2	State FE b/mean a	17.3 [3.453, 50.16]	1000/1000	0
L=2	State FE median a	0.01751 [0.01024, 0.03013]	1000/1000	0
L=2	State FE b/median a	21.67 [3.984, 60.15]	1000/1000	0
L=2	State FE row-weighted a	0.02061 [0.013, 0.03107]	1000/1000	0
L=2	State FE row-weighted ratio	18.41 [3.819, 48.17]	1000/1000	0
L=2	State FE count-weighted a	0.01542 [0.0102, 0.02293]	1000/1000	0
L=2	State FE count-weighted ratio	24.61 [5.111, 57.08]	1000/1000	0
L=2	True D_t b	0.3691 [0.1082, 0.612]	1000/1000	0.001
L=2	True D_t mean a	0.02084 [0.01168, 0.03555]	1000/1000	0
L=2	True D_t ratio	17.71 [3.26, 51.13]	1000/1000	0.001
L=2	Paired pooled/FE ratio	2.164 [0.964, 8.483]	1000/1000	0
L=3	Pooled a	0.01495 [0.01001, 0.02089]	1000/1000	0
L=3	Pooled b	0.56 [0.3851, 0.7603]	1000/1000	0
L=3	Pooled b/a	37.44 [19.39, 68.59]	1000/1000	0
L=3	State FE b	0.3794 [0.111, 0.6669]	1000/1000	0.001
L=3	State FE mean a	0.02192 [0.01247, 0.03692]	1000/1000	0
L=3	State FE b/mean a	17.3 [3.051, 50.11]	1000/1000	0.001
L=3	State FE median a	0.01751 [0.01068, 0.03045]	1000/1000	0
L=3	State FE b/median a	21.67 [3.982, 58.24]	1000/1000	0.001
L=3	State FE row-weighted a	0.02061 [0.01293, 0.0317]	1000/1000	0
L=3	State FE row-weighted ratio	18.41 [3.407, 47.65]	1000/1000	0.001
L=3	State FE count-weighted a	0.01542 [0.01076, 0.02338]	1000/1000	0
L=3	State FE count-weighted ratio	24.61 [4.841, 57.62]	1000/1000	0.001
L=3	True D_t b	0.3691 [0.1072, 0.6252]	1000/1000	0.001
L=3	True D_t mean a	0.02084 [0.0116, 0.03629]	1000/1000	0
L=3	True D_t ratio	17.71 [3.068, 51.41]	1000/1000	0.001
L=3	Paired pooled/FE ratio	2.164 [0.9866, 8.87]	999/1000	0
Point diagnostic	week_only	b=0.5258	No CI	group_gamma_objective_ fallback
Point diagnostic	state_and_week	Unavailable	No CI	iteration_limit

Sample: the R_A specification, n=1748 state-weeks in 51 jurisdictions; the G sample is not used here. Intervals retain signed draws and are conditional on finite estimates when finite/B < 1. Last column gives the nonpositive fraction over all requested draws. FE targets the sampled state occurrences; pooled/FE comparisons are paired within draws. L denotes the within-season week-block length. Week-only and state-and-week fits are point diagnostics; failed fits have no estimate or interval. Pooled rows here use this table's own 1000-draw bootstrap and can differ slightly from the primary 2000-draw interval in Table S1. L=1 pooled: 1000 converged, 0 failed; {}. L=1 state: 999 converged, 1 failed; {"gamma_objective_fallback_ failed": 1}. L=1 state_D: 999 converged, 1 failed; {"gamma_objective_fallback_failed": 1}. L=2 pooled: 1000 converged, 0 failed; {}. L=2 state: 1000 converged, 0 failed; {}. L=2 state_D: 1000 converged, 0 failed; {}. L=3 pooled: 1000 converged, 0 failed; {}. L=3 state: 1000 converged, 0 failed; {}. L=3 state_D: 1000 converged, 0 failed; {}.

Table S5. Centering stages with a common loss normalizer

Statistic / center	Pooled mean	a / change [95% CI]	b [95% CI]	Finite/B
RA / pooled	1.206	0.01495 [0.01016, 0.02083]	0.56 [0.3625, 0.7639]	1000/1000
RA / season	1.206	0.01471 [0.009507, 0.01966]	0.5391 [0.3602, 0.7205]	1000/1000
RA / season_linear	1.206	0.0154 [0.008972, 0.01901]	0.3936 [0.2717, 0.5763]	1000/1000
RA / week	1.206	0.009766 [0.007103, 0.01357]	0.4581 [0.3124, 0.5916]	1000/1000
RA / week_leave_state_out	1.206	0.01016 [0.007654, 0.01481]	0.4841 [0.3492, 0.6636]	1000/1000
RA / week_change	1.206	-0.005189 [-0.01015, -0.0006938]	–	1000/1000
RA / week_reduction_fraction	1.206	34.7 [5.967, 51.9]	–	1000/1000
G / pooled	1.724	0.08401 [0.05163, 0.1222]	1.971 [0.9987, 2.986]	1000/1000
G / season	1.724	0.08163 [0.0504, 0.1156]	1.842 [0.9591, 2.602]	1000/1000
G / season_linear	1.724	0.07864 [0.04808, 0.09897]	1.249 [0.6423, 1.988]	1000/1000
G / week	1.724	0.05607 [0.03962, 0.07812]	1.401 [0.7134, 1.912]	1000/1000
G / week_leave_state_out	1.724	0.05843 [0.04308, 0.08505]	1.477 [0.8089, 2.138]	1000/1000
G / week_change	1.724	-0.02794 [-0.05687, -0.003708]	–	1000/1000
G / week_reduction_fraction	1.724	33.25 [6.805, 51.63]	–	1000/1000

All five centers use the same pooled mean as the relative-loss denominator within each statistic; the denominator is refitted with weights within each bootstrap draw. week_leave_state_out removes the focal state from its weekly center. Reduction is 100*(a_pooled-a_week)/a_pooled, in percent; changes are descriptive and not additive variance components. Intervals here use this table's own 1000-draw bootstrap and can differ slightly from the primary 2000-draw interval in Table S1. RA/pooled: 1000/1000 finite draws; RA/season: 1000/1000 finite draws; RA/season_linear: 1000/1000 finite draws; RA/week: 1000/1000 finite draws; RA/week_leave_state_out: 1000/1000 finite draws; G/pooled: 1000/1000 finite draws; G/season: 1000/1000 finite draws; G/season_linear: 1000/1000 finite draws; G/week: 1000/1000 finite draws; G/week_leave_state_out: 1000/1000 finite draws.

Table S6. Conditional calibration across all declared scenarios

Scenario / statistic	v [95% CI]	Finite/B	Nonpositive / nonfinite	Outside / slope <=0	Failed fits / all
delay_long / RA	0.04587 [0.02996, 0.06461]	2000/2000	0/0	0/0	0/1000
delay_long / G	0.04686 [0.02666, 0.06952]	2000/2000	0/0	3/0	0/1000
delay_short / RA	0.03863 [0.02531, 0.05527]	2000/2000	0/0	0/0	0/1000
delay_short / G	0.04161 [0.02382, 0.06314]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0.25_obs1 / RA	0.02882 [0.01937, 0.04107]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0.25_obs1 / G	0.02967 [0.01709, 0.04456]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0.25_obs1.25 / RA	0.02844 [0.01853, 0.03995]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0.25_obs1.25 / G	0.0294 [0.01681, 0.04331]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0_obs1 / RA	0.03178 [0.02085, 0.04488]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0_obs1 / G	0.03369 [0.01924, 0.05045]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0_obs1.25 / RA	0.03128 [0.02019, 0.04538]	2000/2000	0/0	0/0	0/1000
joint_gi3.2_pn0_obs1.25 / G	0.03348 [0.01895, 0.05184]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0.25_obs1 / RA	0.03764 [0.02444, 0.05359]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0.25_obs1 / G	0.03902 [0.02247, 0.05689]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0.25_obs1.25 / RA	0.03638 [0.02354, 0.0519]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0.25_obs1.25 / G	0.03785 [0.0216, 0.05704]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0_obs1 / RA	0.04002 [0.02558, 0.05767]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0_obs1 / G	0.04282 [0.02407, 0.06368]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0_obs1.25 / RA	0.04035 [0.02648, 0.0569]	2000/2000	0/0	0/0	0/1000
joint_gi3.6_pn0_obs1.25 / G	0.04356 [0.02576, 0.064]	2000/2000	0/0	0/0	0/1000
k01 / RA	0.04023 [0.02553, 0.05736]	2000/2000	0/0	0/0	0/1000
k01 / G	0.04289 [0.02475, 0.0645]	2000/2000	0/0	0/0	0/1000
matched32_binomial / RA	0.03108 [0.02037, 0.04399]	2000/2000	0/0	0/0	0/1000
matched32_binomial / G	0.03287 [0.0186, 0.0489]	2000/2000	0/0	0/0	0/1000
matched32_source_sd21_binomial / RA	0.02884 [0.01908, 0.04027]	2000/2000	0/0	0/0	0/1000
matched32_source_sd21_binomial / G	0.031 [0.01794, 0.04533]	2000/2000	0/0	0/0	0/1000
matched36_binomial / RA	0.04 [0.02623, 0.0562]	2000/2000	0/0	0/0	0/1000
matched36_binomial / G	0.04298 [0.02466, 0.06284]	2000/2000	0/0	0/0	0/1000
national_persistence25 / RA	0.0369 [0.0238, 0.05109]	2000/2000	0/0	0/0	0/1000
national_persistence25 / G	0.03817 [0.02185, 0.05643]	2000/2000	0/0	0/0	0/1000
national_share0 / RA	0.03935 [0.0255, 0.05576]	2000/2000	0/0	0/0	0/1000
national_share0 / G	0.0416 [0.02393, 0.06162]	2000/2000	0/0	0/0	0/1000
national_share65 / RA	0.04014 [0.02576, 0.05685]	2000/2000	0/0	0/0	0/1000
national_share65 / G	0.04327 [0.02461, 0.06553]	2000/2000	0/0	0/0	0/1000
reporting_shock / RA	0.01954 [0.006213, 0.0358]	2000/2000	0/0	0/0	0/1000
reporting_shock / G	0.02903 [0.0117, 0.04918]	2000/2000	0/0	0/0	0/1000
rho_h005 / RA	0.03991 [0.02639, 0.05628]	2000/2000	0/0	0/0	0/1000
rho_h005 / G	0.04283 [0.02548, 0.06297]	2000/2000	0/0	0/0	0/1000
rho_h02 / RA	0.04029 [0.02632, 0.05694]	2000/2000	0/0	0/0	0/1000
rho_h02 / G	0.04317 [0.02494, 0.06466]	2000/2000	0/0	0/0	0/1000
state_persistence50 / RA	0.03244 [0.02095, 0.04559]	2000/2000	0/0	0/0	0/1000
state_persistence50 / G	0.03137 [0.01788, 0.0467]	2000/2000	0/0	0/0	0/1000
true2.2_analyst36 / RA	0.0146 [0.009442, 0.02045]	2000/2000	0/0	0/0	0/1000
true2.2_analyst36 / G	0.01473 [0.008358, 0.02141]	2000/2000	0/0	0/0	0/1000
true2.9_analyst36 / RA	0.02551 [0.0168, 0.03572]	2000/2000	0/0	0/0	0/1000
true2.9_analyst36 / G	0.02679 [0.01524, 0.0401]	2000/2000	0/0	0/0	0/1000
true4.3_analyst36 / RA	0.05807 [0.03763, 0.08382]	2000/2000	0/0	27/0	0/1000
true4.3_analyst36 / G	0.06312 [0.0366,	2000/2000	0/0	78/0	0/1000

Table S7. Model-conditional design size and power

Seasons	Count multiplier	Size [95% CI]	Power [95% CI]	Valid C/N/A	Size compatible
4	1	0.058 [0.03918, 0.08224]	0.996 [0.9856, 0.9995]	500/500/500	yes
4	2	0.06 [0.04084, 0.08455]	0.998 [0.9889, 0.9999]	500/500/500	yes
4	4	0.052 [0.03425, 0.07527]	0.994 [0.9826, 0.9988]	500/500/500	yes
8	1	0.058 [0.03918, 0.08224]	1 [0.9926, 1]	500/500/500	yes
8	2	0.038 [0.02303, 0.05871]	1 [0.9926, 1]	500/500/500	yes
8	4	0.082 [0.05949, 0.1096]	1 [0.9926, 1]	500/500/500	no
12	1	0.058 [0.03918, 0.08224]	1 [0.9926, 1]	500/500/500	yes
12	2	0.07 [0.04924, 0.09601]	1 [0.9926, 1]	500/500/500	yes
12	4	0.054 [0.03588, 0.0776]	1 [0.9926, 1]	500/500/500	yes
20	1	0.032 [0.0184, 0.05145]	1 [0.9926, 1]	500/500/500	yes
20	2	0.058 [0.03918, 0.08224]	1 [0.9926, 1]	500/500/500	yes
20	4	0.056 [0.03753, 0.07992]	1 [0.9926, 1]	500/500/500	yes

Conditional discrimination simulations recycle the observed season/count design. Calibration and evaluation samples are independent. Valid counts are calibration/null/alternative. The final column states whether the size interval contains nominal alpha; all cells are reported, including failed size checks. These simulations do not imply a universal season threshold or a universal 80% power claim. alpha=0.05; replicates/stage=500.

Table S8. Season-specific fits and reporting participation sensitivity

Statistic / period	n	a / change [95% CI]	b / original a	b/a / adjusted a	Finite/B
RA / 2022-23	365	0.01369 [0.0071, 0.02102]	0.6967	50.91	1000/1000
RA / 2023-24	499	0.006174 [0.002667, 0.01542]	0.7018	113.7	1000/1000
RA / 2024-25	356	0.01268 [0.005687, 0.02276]	0.5826	45.93	1000/1000
RA / 2025-26	528	0.02253 [0.008432, 0.03509]	0.3533	15.68	1000/1000
RA / reporting adjustment	1748	0.0002939 [-0.0003183, 0.0008476]	0.01495	0.01466	1000/1000
G / 2022-23	367	0.07179 [0.03786, 0.1162]	2.702	37.64	1000/1000
G / 2023-24	503	0.03046 [0.0118, 0.07462]	2.382	78.2	1000/1000
G / 2024-25	357	0.06469 [0.03191, 0.1278]	2.082	32.19	1000/1000
G / 2025-26	529	0.1383 [0.03715, 0.2252]	0.7411	5.359	1000/1000
G / reporting adjustment	1756	0.000373 [-0.003559, 0.003454]	0.08401	0.08364	1000/1000
Summer 2023	772	15.55	94.24	0.4873	–
Summer 2024	400	17.12	45.59	0.6683	–
Summer 2025	839	16.25	92.31	0.4882	–

Season rows report a with its interval and b, b/a as point estimates only. Reporting-adjustment rows use the a column for original-minus-adjusted a [95% CI], b for original a, and b/a for adjusted a; the original normalizer is fixed. Summer rows use these columns for median count scale, median reporting participation (%) and IQR(log G). The single voluntary summer is descriptive, not a causal or independent replicated intervention. RA/2022-23: 1000/1000 finite draws; RA/2023-24: 1000/1000 finite draws; RA/2024-25: 1000/1000 finite draws; RA/2025-26: 1000/1000 finite draws; G/2022-23: 1000/1000 finite draws; G/2023-24: 1000/1000 finite draws; G/2024-25: 1000/1000 finite draws; G/2025-26: 1000/1000 finite draws.